

Research article

Z-number based fuzzy MCDM models for analyzing non-traditional security threats to finance supply chains: A case study from Vietnam

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ARTICLE INFO

Keywords:

Non-traditional security
Supply chain finance
MCDM
Z number
Fuzzy sets

ABSTRACT

Non-traditional security (NTS) threats have a vast and profound impact on many aspects of economic, political, social, and many other areas, especially supply chain finance (SCF), particularly in countries like Vietnam, which potentially affects the economic efficiency of businesses' supply chain financial, thereby affecting the general economy of the country and the world. In order to prevent and minimize the negative impacts caused by NTS threats to SCF, this study was conducted to identify NTS threats affecting SCF in Vietnam, at the same time calculate the weight of the impact level and find out the cause and effect relationship between them. Solution strategies are also proposed and ranked, thereby serving as a reference basis for relevant parties to choose appropriate response solutions. Due to the multi-criteria nature of NTS threats, the multiple criteria decision-making (MCDM) method is used in combination with the Z-number concept and Fuzzy set theory to approach the problem of certainty and increase the accuracy of study. The NTS threats are first identified through a literature review and then validated for suitability using the DELPHI technique (DELPHI). Suitable threats will be determined by relationship, weighted by Decision Making Trial And Evaluation Laboratory (DEMATEL) method. Proposed strategies are ranked using the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) method. The results indicate that there are 19 NTS factors affecting SCF in Vietnam, and the global economic downturn, pandemic and health crisis, financial crisis and cybersecurity risk are the four root cause factors with the most decisive influence. Businesses and concerns need to prioritize addressing these four threats because they not only have a strong impact but also entail many other threats. The two strategies considered to be the most effective are a sustainable practice and a risk-hedging strategy. Businesses, governments, and stakeholders also should pay attention to the macroeconomic environment, technology, and environment and build sustainable businesses, regularly monitoring economic fluctuations and creating plans to prevent risks.

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<https://doi.org/10.1016/j.heliyon.2024.e31615>

Received 28 January 2024; Received in revised form 17 May 2024; Accepted 20 May 2024

Available online 21 May 2024

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1. Introduction

The Covid-19 pandemic has significantly impacted the global financial system and disrupted the supply chain. Social distancing measures in 2019 led to economic slowdown. In Vietnam, these measures profoundly affected the national economy, with a sharp decline in 2020 economic growth, as reported by the General Statistics Office [1]. Siti Nurhasanah and Ahmad Bahrul Anshori [2] suggest that the COVID-19 pandemic is not essentially a traditional security challenge but a non-traditional security threat. In contrast to traditional security, non-traditional security emerged in the post-Cold War era, focusing on its non-traditional security dimensions. Non-traditional security does not center around states; instead, it concentrates on human security [3]. Non-traditional security argues that threats such as environmental, political, economic, and social issues endanger individuals' lives and properties more than the state's survival [3,4]. Similar to the Covid-19 pandemic, it has caused a series of significant disruptions in the global financial market. Lockdown measures and social distancing have severed the global financial supply chain, posing considerable challenges for businesses and industries.

More and Basu [5] stated that SCF is the management, planning, and control of all transaction activities and processes pertaining to the cash flow among supply chain (SC). Any factor that affects the management, planning, and control of all transaction activities in a business has an impact on supply chain finance. Amidst the backdrop of the COVID-19 pandemic—an NTS threat—measures such as social distancing and lockdowns have significantly altered global banking activities, financial management, operational procedures, and trade, consequently impacting SCF. Beyond COVID-19, risks stemming from other epidemics, as well as environmental, economic, legal, political, and social factors, all constitute NTS threats that profoundly affect SCF in Vietnam. It's imperative for stakeholders to comprehensively grasp the potential impact of these threats in order to devise appropriate plans to mitigate and minimize risks.

To improve supply chain efficiency and provide strategies for the risks posed by non-traditional security challenges affecting the financial supply chain, businesses, governments, and other stakeholders need to recognize, identify, and understand NTS threats. However, previous studies mainly focused on the general risks of supply chains and financial supply chains, as well as factors affecting financial supply chains [6–11]. Previous studies did not mention the concept of non-traditional security in quantifying risks. Given the current world context with many non-traditional security challenges, focusing on identifying non-traditional security risks affecting the financial supply chain is necessary.

To approach this issue, this study was conducted based on the following research questions.

- (1) What non-traditional security challenges affect the financial supply chain in Vietnam?
- (2) What is the cause-and-effect relationship between those non-traditional security challenges, and where is the root cause factor?
- (3) What is the impact of the proposed strategies on those non-traditional security challenges, and what is the priority of those strategies?

In this research, MCDM method combined with Z-Number is used to collect and analyze experts' assessments more accurately in a context where uncertainties and incomplete information exist (P. H. [12]). The MCDM method is well-suited for analyzing and determining the impact of NTS threats and ranking solutions due to the multi-criteria nature of the problem. With numerous NTS threats affecting SCF and various evaluation criteria for ranking strategies, MCDM offers appropriate analytical tools such as DELPHI, DEMATEL, and TOPSIS. These methods are adept at handling complex relationships between factors and differing influence weights, which would pose challenges for other approaches. Furthermore, the integration of Z-number and Fuzzy set theory enhances the accuracy and certainty of data, acknowledging the inherent uncertainty in decision-making processes in real-world scenarios. The research includes three phases: the first phase is identifying and confirming threats - using the Z-Fuzzy Delphi method; the second phase is determining the cause-and-effect relationship between factors and weighting the level of influence - using the Z-Fuzzy Dematel method; the third phase is ranking the proposed strategies - using the Z-Fuzzy Topsis method.

In practical terms, this study aims to comprehensively identify and understand non-traditional security threats pertinent to businesses, governments, and stakeholders, potentially affecting supply chain finance. By assessing threat weights and identifying root cause factors, stakeholders can pinpoint critical areas requiring attention. These insights empower entities to develop proactive strategies for anticipating, preventing, and mitigating security challenges, thus bolstering the resilience of supply chain finance and enhancing overall economic efficiency. Furthermore, proposing and ranking solutions based on effectiveness criteria provides actionable recommendations for governments and businesses to address non-traditional security threats effectively. Previous studies predominantly focused on analyzing risks across the supply chain; however, delving deeper into the perspective of supply chain finance and the concept of non-traditional security offers a more granular understanding, facilitating more targeted and efficient resolution of these emerging threats. Theoretically, the integration of Z-number theory with Fuzzy Number and the MCDM method addresses the issue of certainty levels in expert responses, enhancing the accuracy and reliability of the data and results. Leveraging the DELPHI method to validate variables ensures the relevance and importance of identified threats, thereby contributing to the certainty, specificity, and accuracy of the study outcomes.

The remainder of this research comprises 4 sections. Section 2 delves into foundational concepts, reviews prior research, examines methods, and addresses non-traditional security factors impacting the financial supply chain, proposing strategies. Section 3 introduces the Z-number concept and outlines the implementation steps of the Z-Fuzzy Delphi method, Z-Fuzzy Dematel, and Z-Fuzzy Topsis. Section 4 presents findings and discussions, while section 5 engages in a comprehensive conclusion.

2. Literature review

2.1. Non-traditional security threats

Global scholars are extensively engaged in analyzing and researching non-traditional security challenges due to their profound impact on the economic, political, and social dimensions worldwide. As Caballero Anthony [3] defines, non-traditional risks encompass non-military threats that cannot be addressed through military measures, spanning areas such as resource security and economic warfare. In the context of Vietnam, Nghiem and Nguyen [13] emphasize that non-traditional security is a multifaceted concern, intertwining aspects of food security, environmental considerations, and national security. From a legal standpoint, Nghiem and Nguyen [13] assert that non-traditional security is integral to Vietnam's security and national defense, positioning the safeguarding of national security as a strategic imperative. The non-traditional security threats faced by Vietnam encompass challenges to national independence, sustainable development, socio-political stability, sovereignty, and security [13]. These threats, which include cyber security, currency risks, natural resources, environmental issues, and transnational crime, permeate various sectors, including politics, economics, society, environment, and technology.

2.2. Supply chain finance

Modern supply chains are of increasing interest to governments and businesses globally, with financial considerations standing out as a significant concern. According to Hofmann [14], supply chain finance involves collaborative value creation among two or more organizations within a supply chain, including external service providers, through coordinated planning, direction, and control measures. Pfohl and Gomm [15] describe SCF as a strategy that optimizes capital mobilization by integrating financing processes with customers, suppliers, and service providers, thereby enhancing overall value. Liu et al. [16] characterize SCF as an innovative financial strategy facilitating connections between companies facing supply chain challenges and banks, minimizing the risk of financial imbalances, and adding value to the supply chain. Beyond its financial aspect, Moretto et al. [17] propose that SCF is a fusion of technology strategies and financial services, tightly integrated into the supply chain to enhance efficiency by preventing cost variations, delivery delays, and cash flow disruptions. In a comprehensive view, this study adopts the definition of financial supply chain within the supply chain framework by More and Basu [5], defining SCF as supply chain stakeholders' management, planning, and control of transaction activities and processes related to cash flow to enhance working capital.

Recognizing the critical role of the SCF as the lifeblood of businesses [18], it is crucial to address potential risks that could lead to difficulties or even bankruptcy. In theoretical terms, Chen and Hu [19] highlighted the significance of SCF, demonstrating the importance of financial flows within the supply chain. The three types of flows in the supply chain model indicated that financial flows both forward and backward direction between suppliers and customers, encompass activities such as rebates, financing, consignment, and price protection in the positive direction, and deposits, payments, and exchanges in the negative direction (X. [19]). Therefore, SCF is crucial for maintaining the smooth and seamless operation of the supply chain, as it ensures the connectivity between various links within the supply chain (X. [19–21]). However, when applying this model to risk assessment in SCF, it primarily addresses endogenous risks within the supply chain, overlooking external factors such as non-traditional security challenges that can impact financial flows and SCF. While traditional risks in SCF can be anticipated and managed, non-traditional risks, such as political instability, policy changes, or technological risks [22], pose a challenge due to their unpredictable nature. Accurate, objective, and multi-dimensional analysis and assessment of these non-traditional risks are essential to formulate appropriate response measures.

2.3. Literature review on related research

In contemporary research, various methods, such as descriptive statistics and MCDM, are available to researchers to support their investigations. Descriptive statistics utilize a probability model to describe data, identify classifications, validate results, compare

Table 1

A summary of related works.

No.	References	Methods	Key points
1	Karmaker et al. [35]	Pareto analysis and Fuzzy MICMAC	"Analyzing supply chain risk factors in the small and medium enterprises."
2	Babu and Yadav [36]	FRI and FPPI	"Supply chain risk assessment index"
3	Ul Islam et al. [10]	AHP-TOPSIS	"Assessment of challenges and problems in supply chain among retailers"
4	Magableh and Mistarihi [34]	ANP-TOPSIS	"Evaluate supply chain strategies."
5	Khan et al. [39].	Fuzzy-AHP	"Prioritizing the risks"
6	Kumar and Kumar Barua [7]	DEMATEL	"The interaction among risk factors of the sustainable petroleum supply chain"
7	Akter et al. [40]	Grey-DEMATEL	"The disruption risk factors in the Emergency Life-Saving Drugs supply chains"
8	Moktadir et al. [8]	Pareto analysis and BWM	"Analysis of risk factors in sustainable supply chain management of leather industry"
9	Lahane and Kant [41]	PF-AHP and PF-VIKOR	"Evaluation and ranking of strategies to mitigate circular supply chain risks"
10	Wang et al. [42]	Statistics method	"Drivers and outcomes of supply chain finance adoption"
11	Mzougui et al. [9]	AHP - Fuzzy DEMATEL	"Assessing Supply Chain Risks in the automotive industry"
12	Beka Be Nguema et al. [6]	Statistics method, using SEMPLS	"Factors influencing the adoption of supply chain finance in supply chain effectiveness"
13	Tseng et al. [43]	Fuzzy TOPSIS	"Sustainable supply chain finance factor"
14	Wan Ahmad et al. [11]	BWM	"External forces affecting the sustainability of oil and gas supply chain"

schemes, and optimize classifiers through a probabilistic search strategy [23]. Despite its utility, this method presents limitations, including potential bias and inaccuracy in statistical data, challenges in establishing causal relationships, and issues of data accessibility and potential misuse [24]. In contrast, MCDM stands out as a comprehensive and exploratory method, facilitating the exploration of factor interactions within the research context, including ranking factors [25]. Given its richness and completeness, this research applies using the MCDM method.

Nowadays, the financial supply chain is gaining increased attention from government entities and businesses, prompting a heightened focus on assessing its impacts and associated risks within organizations [26]. Managing risks in supply chain finance poses a substantial challenge. Recognizing the complexity of analyzing non-traditional risks in SCF, researchers advocate for the MCDM approach [27]. Thakkar [25] highlights the comprehensive and versatile nature of MCDM, applicable to both personal and professional problem-solving scenarios. MCDM theory operates on the premise that decision-makers choose from alternatives with known target values or attributes, considering potential risks or uncertainties. When applying MCDM in specific contexts, considerations include comparing criteria, assessing trade-offs, and navigating constraints in decision-making [28]. Notable MCDM methods encompass Technique for Order Preference by Similarity to Ideal Strategies (TOPSIS) [29], Decision Making Trial and Evaluation Laboratory (DEMATEL) [9], Delphi Technique (DELPHI) [30], Simple Additive Weightage (SAW) [31], Analytic Hierarchy Process (AHP) [10], Multi-objective Optimization based on Ratio Analysis Method (MOORA) [32] and Analytic Network Process (ANP) [33], Best Worst Method (BWM) [32].

Table 1 has an overview of MCDM Methods in various topics related to supply chain and supply chain finance. Magableh and Mistarihi [34] employed a combination of ANP and TOPSIS to assess COVID-19's impact on the financial supply chain. Karmaker et al. [35] contributed to the discourse by analyzing small and medium risks using Pareto analysis and Fuzzy Micmac methods. Within the supply chain domain, an extensively explored field, authors such as Karmaker et al. [35] and Babu and Yadav [36] have integrated MCDM techniques with Pareto analysis, Fuzzy MICMAC, FRI, and FPII. The application of these methodologies serves to evaluate and analyze various aspects of supply chain management comprehensively. In terms of combining methods, it is worth noting that Shao et al. [37] proposed a model for assessing smart city sustainable development that assessed the Chinese smart city model of Xiamen by combining the Z-Fuzzy number with TOPSIS and DEMATEL methods. This approach enhances result certainty and analysis reliability, offering a valuable foundation for managerial decision-making [38].

While the use of MCDM methods proves invaluable in research and managerial decision-making, it is noteworthy that many researchers tend to employ individual or limited combinations of methods. Notably, Abdel-Basset et al. [44] applied TOPSIS and BWM for sustainable supply chain finance sustainability. Chen et al. [45] utilized TOPSIS to evaluate a supply chain finance platform. However, certain studies indicate uncertainties and reliability concerns in their results. Therefore, adopting a comprehensive approach by hybrid the Z-Fuzzy number concept with the MCDM method to access information about their level of certainty when experts assess the survey enhances accuracy and provides a more complete representation of desired outcomes in the field of supply chain finance [37].

The escalating challenges in Vietnam's financial supply chain amidst the country's development have been a focal point. Despite this, the SCF system remains relatively underexplored in research. Recognizing the gaps left by previous studies, such as the lack of attention to the topic and the absence of integration of MCDM methods specific to Z numbers, our research aims to address these limitations. The distinctive approach revolves around leveraging the power of Z-numbers and combining this methodology with other established methods. Z-numbers, characterized by their limiting and reliability components, emerge as a robust tool for articulating decision information [46,47]. The inherent strengths and advantages of Z-numbers have garnered global interest, prompting researchers to delve deeper into expanding their theoretical framework and applications, particularly within the realm of fuzzy set theory and its implications in decision-making. The current trajectory of Z-number research reflects a growing fascination with fuzzy set theory, emphasizing its practicality in decision-making under uncertainty. Within the domain of MCDM, Z-numbers come into play by accounting for the certainty of information and evaluating decisions in uncertain scenarios. Researchers often intertwine Z-numbers with probability and fuzzy logic theories, employing methodologies like DEMATEL and TOPSIS [48]. Our research pioneers a novel approach by synergizing three key methods: DELPHI, DEMATEL, and TOPSIS, while integrating the concept of Z-numbers. Specifically, the DELPHI method validates the suitability of threats through expert assessment, with only those surpassing a predefined threshold proceeding to subsequent phases. DEMATEL calculates threat weights and discerns cause-and-effect relationships to identify critical root causes. Finally, the TOPSIS method ranks solutions based on threat importance and their corresponding weights determined by DEMATEL. This multifaceted approach allows for a more precise and objective exploration of financial supply chain challenges in Vietnam. By employing this combined methodology, we aim to offer financial supply chain managers in Vietnam comprehensive, accurate, and constructive strategies to effectively navigate and mitigate risks in their operations.

2.4. Literature review on NTS threats affecting SCF

Conducting a comprehensive Political, Economic, Social, Technological, Legal, and Environmental (PESTEL) analysis has revealed non-traditional security challenges with the potential to significantly impact the financial supply chain. The PESTEL model is a renowned framework in strategic analysis, commonly employed to assess external factors impacting business operations [49]. Given that NTS threats directly influence an enterprise's supply chain finance SCF, it is well-suited for analysis using the PESTEL model. Thus, applying the PESTEL framework enables the definition, identification, and classification of NTS threats, thereby enhancing the efficacy of literature review, synthesis, and problem generalization. To initiate this analysis, an exhaustive search was executed through prominent library databases such as Science Direct, Scopus, and Web of Knowledge. Employing multiple keywords and strings such as "supply chain finance," "supply chain threats," "supply chain risks," and "non-traditional security," the search targeted both abstracts and the main body of scholarly papers. Major journals in logistics, supply chain management, finance, and management were

Table 2

List of NTS threats potentially affecting SCF.

Threats	References
THR1	Political Sanctions [22,50]
THR2	Political Instability [22]
THR3	Legal and Regulatory Changes [22,51]
THR4	Trade war and economic sanction [22,50]
THR5	Global Economic Downturn [22]
THR6	Trade Disputes [36,52]
THR7	Inflationary Pressures [40,53]
THR8	Transnational commercial crimes [50,51]
THR9	Currency Exchange Rate Fluctuations [50,53]
THR10	Financial crises [51,54]
THR11	Social Unrest [51]
THR12	Human Rights Violations [40,50,51,53]
THR13	Terrorism [22,50]
THR14	Cultural Differences and Communication Risks [8,55]
THR15	Migration and demographic changes [51]
THR16	Cybersecurity Risks [7,53]
THR17	Technology Obsolescence [8]
THR18	Data Privacy Concerns [36]
THR19	Transportation and other logistics risk [40,36,52]
THR20	Pandemics and Health Crises [51,54]
THR21	Climate Change and Natural Disasters [8,51]
THR22	Natural Resource Scarcity [53]

THR1 – Political Sanctions: Sanctioning a country will impact trade, disrupt financial operations, and overall impact SCF's stability [22].

THR2 – Political Instability: Expressed through protests or constant changes of the ruling class and creates changes that affect economic policy and financial stability. Political instability leads to thiomersal development with its variable, it is the principal determinant of primary differences in financial development [22].

THR3 – Legal and Regulatory: Laws and regulations significantly impact financial compliance in the supply chain [51]. This causes challenges in terms of increasing legal loopholes that unintentionally create negative impacts on SCF's operating environment.

THR4 – Trade War and Economic Sanctions: Conflicts between countries related to economic issues lead to the implementation of mutual sanctions, thereby creating a trade war [51]. This disrupts financial operations and affects the overall stability of SCF.

THR5 – Global Economic Downturn: Negative indicators in economic activities lead to reduced spending and economic recession [22]. The global economic recession puts great pressure on the financial system, increasing the possibility of default and disrupting overall financial stability.

THR6 - Trade Disputes: Disagreements between countries over trade policies lead to changes in tariffs that create changes in the overall cost structure of financial supply chain operations [36].

THR7 - Inflationary Pressures: Inflation pressure directly impacts the country's purchasing power and economic stability [40]. It could increase costs and prices of goods and services, negatively affecting SCF's overall finances and stability.

THR8 - Transnational Commercial Crimes: Illegal activities of a cross-border nature, such as fraud, smuggling, and money laundering, cause revenue loss, negatively affecting business operations, which can lead to a weakening of the economy and financial supply chain [51].

THR9 - Currency Exchange Rate Fluctuations: Fluctuations in currency values affect transactions and financing costs. Exchange rate fluctuations affect commodity prices, changing the financial structure of SCF [53].

THR10 - Financial Crises: Severe financial market disruption could lead to an economic crisis. The sudden loss of most of the nominal value of financial assets creates pressure and directly affects the operations of SC and SCF [51].

THR11 – Social Unrest: Social discontent leads to widespread protests. Social unrest can disrupt and paralyze logistics, affecting goods' timely movement [51].

THR12 – Human Rights Violations: Actions that violate fundamental rights and freedoms, causing human rights violations. Actors involved in human rights abuses may experience reputational deterioration affecting their overall standing in the SCF [53].

THR13 – Terrorism: Terrorist acts threaten policies and infrastructure systems, disrupting supply chain operations and financial transactions and reducing the safety and reliability of SCF transactions [36].

THR14 – Cultural Differences and Communication Risks: Differences in cultural norms and communication styles affect cooperation, creating challenges in communication, creating ineffective foreign affairs, and affecting the overall effectiveness of SCF [55].

THR15 - Migration and Demographic Changes: Migration and changes in demographic structure directly impact the labor force. These changes create consumer demands and require improvements in the supply chain that impact the predictability of the consumer balance and the stability of SCF operations [51].

THR16 - Cybersecurity Risks: Cyber-attacks pose a threat to IT systems. These threats compromise the trustworthiness and security of financial transactions, changing the security paradigm between transactions [7].

THR17 - Technology Obsolescence: Technology becomes obsolete due to scientific and technical progress. The rapid

obsolescence of technology may require a large enough additional cost in the expenditure structure to carry out frequent upgrades, reducing the financial efficiency of the SCF [8].

THR18 - Data Privacy Concerns: Threats of intrusion and unauthorized use of personal or financial information create data privacy concerns contributing to increased surveillance and security practices. Besides, it also creates potential legal challenges, affecting SCF's overall compliance [36].

THR19 – Transportation and Other Logistics Risk: Breakdowns in logistics due to supplier errors or geopolitical issues leading to disruptions in the supply chain can lead to payment delays and increased credit risk, affecting SCF's liquidity and overall stability [52].

THR20 - Pandemics and Health Crises: Outbreaks of infectious diseases impair public health and economic activity. This situation leads to a workforce shortage, affecting production capacity and thus affecting SCF operations due to the stagnation of supply chain activities [51].

THR21 - Climate Change and Natural Disasters: Extreme weather events and geological impacts impact infrastructure and supply chains. Natural disasters cause damage to infrastructure and disruption of the transportation process, thereby creating unnecessary costs that directly affect SCF [8].

THR22 - Natural Resource Scarcity: Resources such as minerals are limited, affecting manufacturing industries [53]. The scarcity of critical natural resources can lead to additional costs to find alternative raw materials and reduce the efficiency of production and business activities, overall affecting the stability of SCF.

scrutinized, ensuring a comprehensive review of relevant literature. This methodological approach extended beyond international journals, encompassing the exploration of proceedings from leading international conferences and published books. Articles touching on supply chain finance in introductory remarks or as secondary research themes were meticulously discarded to maintain precision and relevance in the review process. Table 2 summarizes 22 NTS threats that can potentially affect SCF that will be used for research in this study.

2.5. Literature review on strategies

Based on a comprehensive literature review, numerous measures have been taken to minimize the impact of non-traditional security challenges on the financial supply chain [22]. Diversifying the market facilitates establishing new cooperative relationships with suppliers and customers from many different countries to limit dependence on a single market, create diversity in foreign revenue, and limit the impact of sanctions [56]. Compliance and regulatory update information to ensure compliance with domestic and foreign regulations to minimize serious violations, leading to the need to adjust current regulations [57]. It is also necessary to develop risk-prevention strategies to deal with unwanted events that may suddenly occur [58]. These activities also require the active participation of the local community to take advantage of social resources, reduce the risk of social unrest, and strengthen solidarity for joint recovery. Against potential negative impacts on the financial supply chain [59]. Dynamic currency management uses financial tools and strategies to manage currency and exchange rate fluctuations and minimize risk from financial crises [60]. Ethical supply chain practices through promoting ethical supply chain practices to address human rights violations, terrorism, and cultural differences, foster a positive image, and minimize risk-related risks [61]. Strong cyber security measures are implemented through investments in advanced security measures to protect against cyber threats, ensuring the integrity and confidentiality of transactions and data finance [62]. In addition, blockchain technology should also be applied to improve safety, reduce the risk of data tampering, improve transparency, and provide decentralized control. It is also necessary to develop and refine supply chain recovery plans to address the consequences of pandemics, health crises, climate change, and natural disasters [63]. Finally, sustainable measures to solve environmental problems contribute to a long-term vision [64]. Therefore, this study looks at ten potential strategies for non-traditional security challenges to the financial supply chain in Table 3.

3. Proposed method

3.1. Basic concepts of Z-number

Z-Number, introduced by Zadeh [46], is a method used to perform calculations where the information is not in a completely certain and reliable environment. Comprising two integral components, Z-Number and fuzzy information, reflecting both the assessment score and the reliability level associated with the assessment. In contexts where fuzzy events are present, probability and reliability serve as

Table 3
List of potential strategies.

Strategies	References
S1	Diversification of Markets [56,65]
S2	Continuous Monitoring and Compliance [57]
S3	Risk Hedging Strategies [58,66]
S4	Dynamic Currency Management [60]
S5	Community Engagement [59,67]
S6	Ethical Supply Chain Practices [61,68]
S7	Robust Cybersecurity Measures [62]
S8	Apply Block-Chain technology [69]
S9	Supply Chain Resilience Planning [63]
S10	Sustainable Practices [64]

metrics to gauge the level of certainty. Z-numbers adeptly transform data from these metrics into a set of fuzzy numbers, yielding answers in linguistic form and expressed as Z-fuzzy numbers. Today's research underscores the integration of Z-Fuzzy numbers within MCDM frameworks, including methodologies like Z-Fuzzy ANP, TOPSIS, BWM, and DEMATEL [37,70]. This combination enhances the precision of decision-making processes in uncertain environments. To elucidate this concept further, the following steps outline the application of Z-Number in conjunction with MCDM methodologies.

Z-number is presented as $Z = (\tilde{F}, \tilde{R})$, with $\tilde{F}$ using triangular fuzzy numbers for the assessment value, and $\tilde{R}$ is the reliability (level of certainty) of $\tilde{F} = (f, \mu_{\tilde{F}})|x \in [0, 1]$ and $\tilde{R} = (x, \mu_{\tilde{R}})|x \in [0, 1]$. $\tilde{F}$ and $\tilde{R}$ are both triangular membership functions. Using Equation (1) to convert $\tilde{R}$ to crisp value:

$$\alpha = \frac{\int^x \mu_{\tilde{R}} dx}{\int^u \mu_{\tilde{R}} dx} \quad (1)$$

After adding the reliability weight α to the evaluation value $\tilde{F}$, using Equation (2) to calculate the weighted Z-numbers:

$$Z^\alpha = \{(x, \mu_{\tilde{F}^\alpha}) | \mu_{\tilde{F}^\alpha}(x) = \alpha \cdot \mu_{\tilde{F}}(x), x \in \sqrt{\alpha}x\} \quad (2)$$

Assume that there are n criteria in an evaluation system, $c_i = \{c_1, c_2, \dots, c_n\}$. The criteria must be applied in pairwise comparisons to examine the interactions between each criterion, or to gauge how much c_i influences c_j . Table 4 delineates the linguistic assessment scale, including its code and membership function, whereas Table 5 presents the reliability scale along with its corresponding code and membership function [37].

Illustration example: Suppose that we have to transform the Z-number $Z = (\tilde{F} = FI, \tilde{R} = M)$ into a triangular fuzzy number. Firstly, $Z = (\tilde{F} = FI, \tilde{R} = M)$ is converted to $Z = [(0.1, 0.2, 0.3), (0.3, 0.5, 0.7)]$

Using Equation (3), the reliability's membership function is transformed into a crisp score:

$$\alpha = \frac{\int^x \mu_{\tilde{R}} dx}{\int^u \mu_{\tilde{R}} dx} = \frac{\int_{0.3}^{0.5} x \left(\frac{x-0.3}{0.5-0.3} \right) dx + \int_{0.5}^{0.7} x \left(\frac{0.7-x}{0.7-0.5} \right) dx}{\int_{0.3}^{0.5} \left(\frac{x-0.3}{0.5-0.3} \right) dx + \int_{0.5}^{0.7} \left(\frac{0.7-x}{0.7-0.5} \right) dx} = 0.5 \quad (3)$$

Next, α is added to the evaluation result using Equation (4) $\tilde{F} = FI$

$$Z^\alpha = \{(0.1, 0.2, 0.3) | \alpha = 0.5\} \quad (4)$$

Lastly, Equation (5) is used to transform the weighted Z-number into the regular fuzzy number:

$$Z' = (\sqrt{0.5} \cdot 0.1, \sqrt{0.5} \cdot 0.2, \sqrt{0.5} \cdot 0.3) = (0.0707, 0.1414, 0.2121) \quad (5)$$

Incorporating a set of five assessment scores and corresponding levels of certainty, Table 6 presents a comprehensive array of 25 semantic scales. Each scale delineates the assessment alongside its associated level of certainty, providing a nuanced perspective on the evaluations.

The scale of assessments used for the performance evaluation is shown in Table 6. Similarly, apply the previously discussed membership function establishment concept from Z-Fuzzy Dematel to Z-Fuzzy Topsis technology to calculate Z-Fuzzy Topsis's semantic variable and its membership function. The Z-Fuzzy Topsis assessment scale is shown in Table 7. The calculation outcome is presented in Table 8.

3.2. Proposed model of Z-Fuzzy Delphi, Z-Fuzzy Dematel and Z-Fuzzy Topsis

3.2.1. Research process

The proposed model includes 3 phases, respectively Z-Fuzzy Delphi, Z-Fuzzy Dematel, and Z-Fuzzy Topsis. The research process is shown in Fig. 1.

Research begins with the research team defining the goals and research questions. After that, a literature review on the research topic and research framework was conducted to improve methods and propose research models. A literature review on non-traditional security factors affecting the financial supply chain is also conducted to select potentially influential factors. A group of experts was

Table 4
Dematel's code, membership function, and assessment scale.

Linguistic scale	Code	Membership function
Absolute influence	AI	(0.3, 0.4, 0.4)
Very high influence	VI	(0, 0.1, 0.2)
Fair influence	FI	(0.1, 0.2, 0.3)
Weak influence	WI	(0.2, 0.3, 0.4)
Equal influence (No influence)	EI (NI)	(0, 0, 0.1)

Table 5

The expert assessment's membership function, code, and reliability scale.

Linguistic scale	Code	Membership function
Very high	VH	(0.7, 1, 1)
High	H	(0.5, 0.7, 0.9)
Medium	M	(0.3, 0.5, 0.7)
Low	L	(0.1, 0.3, 0.5)
Very low	VH	(0, 0, 0.3)

Table 6

Z-Fuzzy Dematel membership functions and semantic scales.

Assessment Reliability	EI	WI	FI	VI	AI
VL	(0, 0, 0.0316)	(0, 0.0316, 0.0632)	(0.0316, 0.0632, 0.0949)	(0.0632, 0.0949, 0.1265)	(0.0949, 0.1265, 0.1265)
L	(0, 0, 0.0548)	(0, 0.0548, 0.1095)	(0.0548, 0.1095, 0.1643)	(0.1095, 0.1643, 0.2191)	(0.1643, 0.2191, 0.2191)
M	(0, 0, 0.0707)	(0, 0.0707, 0.1414)	(0.0707, 0.1414, 0.2121)	(0.1414, 0.2121, 0.2828)	(0.2121, 0.2828, 0.2828)
H	(0, 0, 0.0837)	(0, 0.0837, 0.1673)	(0.0837, 0.1673, 0.251)	(0.1673, 0.251, 0.3347)	(0.251, 0.3347, 0.3347)
VH	(0, 0, 0.0949)	(0, 0.0949, 0.1897)	(0.0949, 0.1897, 0.2846)	(0.1897, 0.2846, 0.3795)	(0.2846, 0.3795, 0.3795)

Table 7

Topsis's code, membership function, and assessment scale.

Linguistic scale	Code	Membership function
Very Good	VG	(0.8, 0.9, 1)
Good	G	(0.6, 0.7, 0.8)
Fair	F	(0.4, 0.5, 0.6)
Poor	P	(0.2, 0.3, 0.4)
Very Poor	VP	(0, 0.1, 0.2)

Table 8

Z-Fuzzy Topsis membership functions and semantic scales.

Assessment Reliability	VP	P	F	G	VG
VL	(0, 0.0316, 0.0632)	(0.0632, 0.0949, 0.1265)	(0.1265, 0.1581, 0.1897)	(0.1897, 0.2214, 0.253)	(0.253, 0.2846, 0.3162)
L	(0, 0.0548, 0.1095)	(0.1095, 0.1643, 0.2191)	(0.2191, 0.2739, 0.3286)	(0.3286, 0.3834, 0.4382)	(0.4382, 0.493, 0.5477)
M	(0, 0.0707, 0.1414)	(0.1414, 0.2121, 0.2828)	(0.2828, 0.3536, 0.4243)	(0.4243, 0.495, 0.5657)	(0.5657, 0.6364, 0.7071)
H	(0, 0.0837, 0.1673)	(0.1673, 0.251, 0.3347)	(0.3347, 0.4183, 0.502)	(0.502, 0.5857, 0.6693)	(0.6693, 0.753, 0.8367)
VH	(0, 0.0949, 0.1897)	(0.1897, 0.2846, 0.3795)	(0.3795, 0.4743, 0.5692)	(0.5692, 0.6641, 0.7589)	(0.7589, 0.8538, 0.9487)

established to evaluate the relevance and influence of each factor on the financial supply chain and evaluate the appropriateness of the proposed strategies. The team of experts is all working, researching, and teaching in the supply chain/logistics management field.

The identified factors undergo expert evaluation for relevance and significance to the research topic. The Z-Fuzzy Delphi technique is employed to synthesize and validate the results, accepting factors deemed pertinent for the subsequent phase. In Phase 2, the Z-Fuzzy Dematel method is applied. Experts assess the mutual influence levels between factors, establishing cause-and-effect relationships along with the weight of each factor's influence. In phase 3, the proposed strategies are evaluated by experts for their impact on the identified factors. The Z-Fuzzy TOPSIS method is employed, leveraging the weights calculated in Phase 2. This comprehensive approach enables the ranking of strategies ideality. Data analysis is carried out using Microsoft Excel version 2019 and Statistical Package for the Social Sciences (SPSS) software version 26.

3.2.2. Z-Fuzzy Delphi method

The combination of Fuzzy number theory and the traditional DELPHI method was proposed by Ishikawa et al [71]. For better measurement, the Z-Fuzzy number has been combined with the traditional DELPHI method. By adding the concept of the degree of certainty of the evaluation, the integration Z-Fuzzy number and the traditional DELPHI method will be a more accurate method of measuring each expert's assessment of the appropriateness of the factors (P.-H. [72]). The scale for assessing the level of influence and the corresponding level of certainty uses 25 semantic scales in Table 6.

The value of the influence of factor c is assessed by expert e as $w = (d_{ec}; g_{ec}; t_{ec})$, $e = 1, 2, 3, \dots, n$, $c = 1, 2, 3, \dots, m$ then, weight w_c of factor c is $w_c = (d_c; g_c; t_c)$ where $d_c = \min(d_{ec})$, $g_c = \sqrt[n]{\prod_{e=1}^n g_{ec}}$, and $t_c = \max(t_{ec})$.

The following equations are suggested to calculate the convex combination value V_c by α -cut method using Equation (6):

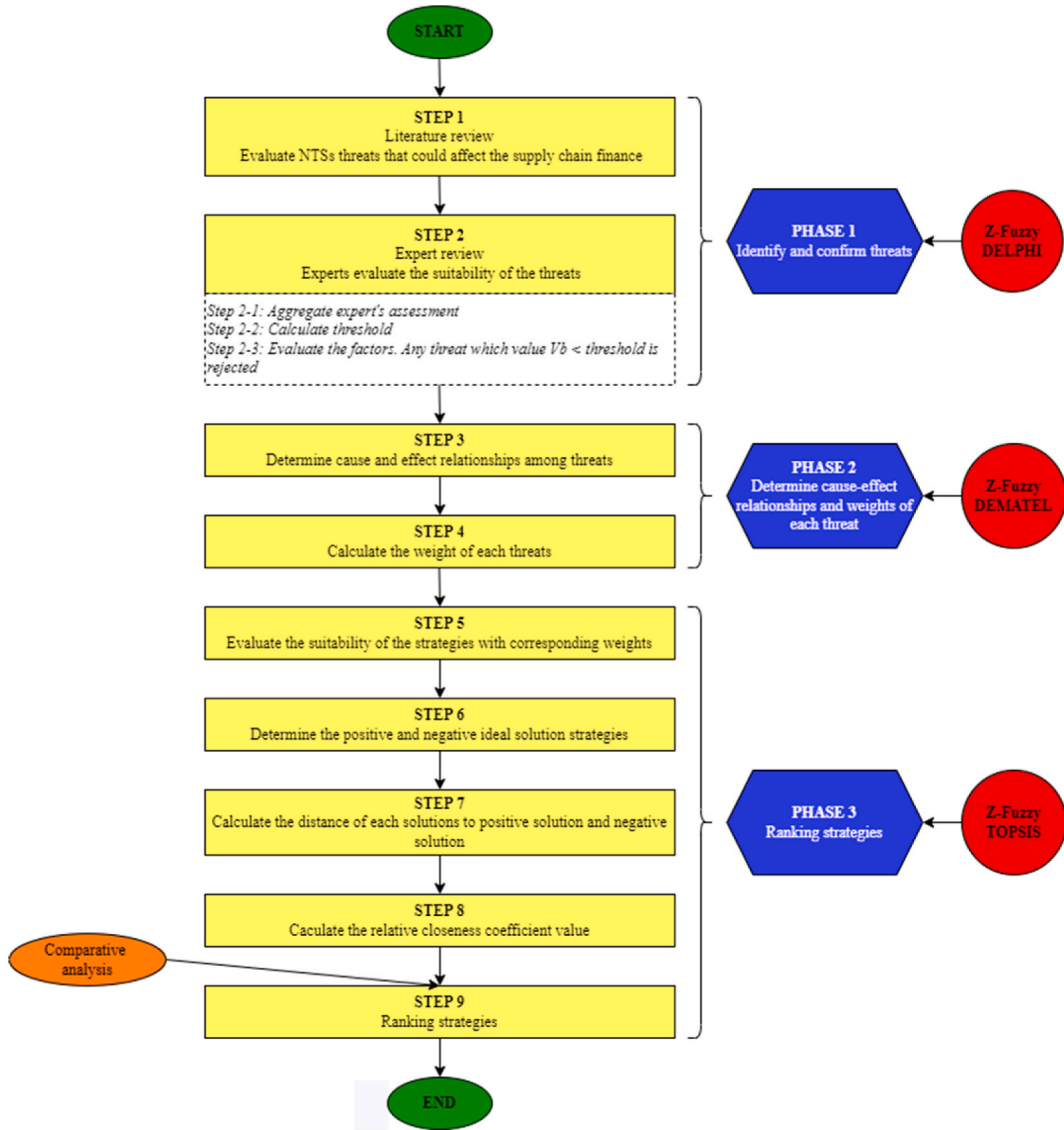


Fig. 1. Research process.

$$\begin{aligned} u_c &= t_c - \varphi \cdot (t_c - g_c) \\ l_c &= d_c - \varphi \cdot (g_c - d_c) \end{aligned} \quad (6)$$

with $c = 1, 2, 3, \dots, m$

Generally, φ is indicated by 0.5 in the typical scenario [73]. Depending on whether the experts are positive or negative perceivers, this value can be anywhere between 0 and 1. The following can be used to generate the exact value of V_c using Equation (7):

$$V_c = \int (u_b, l_b) = \xi \cdot [u_b + (1 - \xi) \cdot l_b] \quad (7)$$

Where ξ is used to characterize a decision maker's positivity level and to create equilibrium between the core opinions of the expert group. Value 0.5 is typically used for ξ .

Next, we calculate threshold ϱ to validate the suitability of the factor using Equation (8). If $V_c \geq \varrho$ factor c is accepted. In contrast, factor c is rejected. Factors that are not rejected will be tested for cause-and-effect relationships and their weights will be calculated using the Dematel method presented below.

$$\varrho = \sum_{c=1}^m \left(\frac{V_c}{m} \right) \quad (8)$$

3.2.3. Z-Fuzzy Dematel

Fontela and Gabus in 1976 first time used the DEMATEL technique in research. Taking into account the opinions of experts, it has been able to resolve numerous worldwide complex issues in the scientific, political, and economic spheres [25]. Using the DEMATEL method will help find the cause-and-effect relationship between factors, thereby finding out what the root cause factor is, and thereby proposing strategies. Besides, using DEMATEL it is also possible to calculate the weight of the influence of each factor on the problem, as input for the Topsis method to rank strategies to the problem. This study combines the Z-Fuzzy number with traditional DEMATEL. This combination will increase the decision group's dependability, reduce the likelihood of information loss during the evaluation process, and improve the assessment's accuracy [37]. Below is the calculation step of Z-Fuzzy Dematel.

Step 1. Create a set of factors for evaluation.

A panel of experts is gathered to ascertain a set of factors $I_i = \{I_1, I_2, \dots, I_n\}$

Step 2. Creating the direct relationship matrix $\otimes H$

It is necessary to assess the influence of n factors. Each expert verifies the results using the reliability scale to ascertain the degree of confidence in them after utilizing the evaluation scale to ascertain the direct impact of criteria i on criteria j . This step is associated with the extraction of the direct relationship matrix. Experts are given a DEMATEL questionnaire with a Z-number to complete in this step.

An optimization model can be used to obtain group judgments using Equation (9) to prevent the extreme opinions of some experts from distorting evaluation results:

$$\begin{aligned} \min z = & \sum_{k=1}^K (d_{ij} - d_{ij}^k)^2 + \sum_{k=1}^K (g_{ij} - g_{ij}^k)^2 + \sum_{k=1}^K (t_{ij} - t_{ij}^k)^2 \\ \text{s.t. } & \begin{cases} \min_k d_{ij}^k \leq d_{ij} \leq \max_k d_{ij}^k \\ \min_k g_{ij}^k \leq g_{ij} \leq \max_k g_{ij}^k \\ \min_k t_{ij}^k \leq t_{ij} \leq \max_k t_{ij}^k \\ d_{ij} \leq g_{ij} \leq t_{ij} \end{cases} \end{aligned} \quad (9)$$

with K is the number of experts and k represents each expert, $k = 1, 2, \dots, K$. The group judgment's minimum, intermediate, and maximum elements are represented by the symbols d_{ij} , g_{ij} , and t_{ij} , respectively. The model's optimal strategies can be determined by using Equation (10) to calculate the partial differentiation d_{ij} , as shown, the same as for g_{ij} and t_{ij} :

$$\begin{aligned} \frac{\partial z}{\partial d_{ij}} = 2 \sum_{k=1}^K (d_{ij} - d_{ij}^k) \cdot 1 = 0 & \implies d_{ij} = \frac{\sum_{k=1}^K d_{ij}^k}{K} \\ g_{ij} &= \frac{\sum_{k=1}^K g_{ij}^k}{K} \\ t_{ij} &= \frac{\sum_{k=1}^K t_{ij}^k}{K} \end{aligned} \quad (10)$$

All expert assessments will be combined into a group direct relationship matrix $\otimes H$ using Equation (11):

$$\begin{aligned} \otimes H = [\otimes h_{ij}]_{n \times n} &= \begin{bmatrix} \otimes h_{11} & \otimes h_{12} & \dots & \otimes h_{1j} & \dots & \otimes h_{1n} \\ \otimes h_{21} & \otimes h_{22} & \dots & \otimes h_{2j} & \dots & \otimes h_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes h_{i1} & \otimes h_{i2} & \dots & \otimes h_{ij} & \dots & \otimes h_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes h_{n1} & \otimes h_{n2} & \dots & \otimes h_{nj} & \dots & \otimes h_{nn} \end{bmatrix}_{n \times n} \\ i=j &= 1, 2, \dots, n \end{aligned} \quad (11)$$

where $\otimes h_{ij} = (h_{ij}^d, h_{ij}^g, h_{ij}^t)$. The matrix's diagonal elements are 0, i.e., $\otimes \alpha_{ij} = 0$.

Step 3. Compute normalized direct relationship matrix $\otimes X$

The value of $\otimes h_{ij}$ in range form 0 to 4, using Equation (12) and (13) to normalize this value into $\otimes x_{ij}$ in 0–1 interval:

$$\otimes X = [\otimes x_{ij}]_{n \times n} = \begin{bmatrix} \otimes \theta \cdot h_{11} & \otimes \theta \cdot h_{12} & \dots & \otimes \theta \cdot h_{1j} & \dots & \otimes \theta \cdot h_{1n} \\ \otimes \theta \cdot h_{21} & \otimes \theta \cdot h_{22} & \dots & \otimes \theta \cdot h_{2j} & \dots & \otimes \theta \cdot h_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes \theta \cdot h_{i1} & \otimes \theta \cdot h_{i2} & \dots & \otimes \theta \cdot h_{ij} & \dots & \otimes \theta \cdot h_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes \theta \cdot h_{n1} & \otimes \theta \cdot h_{n2} & \dots & \otimes \theta \cdot h_{nj} & \dots & \otimes \theta \cdot h_{nn} \end{bmatrix}_{n \times n} \quad (12)$$

where $\otimes x_{ij} = (x_{ij}^d, x_{ij}^g, x_{ij}^t)$.

$$\theta = \min \left\{ \frac{1}{\max_i \sum_{j=1}^n h_{ij}^t}; \frac{1}{\max_i \sum_{i=1}^n h_{ij}^t} \right\} \quad (13)$$

Step 4. Calculate the total influence matrix $\otimes V$.

The total influence matrix $\otimes V$ is calculated by computing the integration of the normalized direct relationship matrix $\otimes X$. Equations (14)–(16) are used in this step to summarize all direct and indirect influence relationships from the power of one to the power of infinity.

$$\otimes V = [\otimes v_{ij}]_{n \times n} = \begin{bmatrix} \otimes v_{11} & \otimes v_{12} & \cdots & \otimes v_{1j} & \cdots & \otimes v_{1n} \\ \otimes v_{21} & \otimes v_{22} & \cdots & \otimes v_{2j} & \cdots & \otimes v_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes v_{i1} & \otimes v_{i2} & \cdots & \otimes v_{ij} & \cdots & \otimes v_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes v_{n1} & \otimes v_{n2} & \cdots & \otimes v_{nj} & \cdots & \otimes v_{nn} \end{bmatrix}_{n \times n} \quad (14)$$

$i, j = 1, 2, \dots, n$

where $\otimes v_{ij} = (v_{ij}^d, v_{ij}^g, v_{ij}^t)$.

$$\otimes V = \otimes X + \otimes X^2 + \dots + \otimes X^\infty \quad (15)$$

$$\begin{aligned} \otimes V &= \otimes X + \otimes X^2 + \dots + \otimes X^\infty \\ &= \otimes X(I + \otimes X + \otimes X^2 + \dots + \otimes X^{\infty-1}) \\ &= \otimes X(I - \otimes X^\infty)(I - \otimes X)^{-1} = \otimes X(I - \otimes X)^{-1} \end{aligned} \quad (16)$$

Where $\otimes X^\infty = [0]_{n \times n}$ and I is the identity matrix

Step 5. Establishing a total influence matrix map (INRM) to identify the mutual influence of development projects.

Firstly, sum up all rows of the total influence matrix $\otimes V$ to calculate $\otimes r$. Similarly, $\otimes c$ is calculated by summing up all rows using Equation (17)–(20):

$$\otimes r = [\otimes r_i]_{n \times 1} = (\otimes r_1, \otimes r_2, \dots, \otimes r_i, \dots, \otimes r_n) \quad (17)$$

$$[\otimes r_i]_{n \times 1} = \left[\sum_{j=1}^n \otimes v_{ij} \right]_{n \times 1} \quad (18)$$

$$\otimes c = [\otimes c_i]_{1 \times n} = (\otimes c_1, \otimes c_2, \dots, \otimes c_j, \dots, \otimes c_n)^T \quad (19)$$

$$[\otimes c_j]_{1 \times n} = \left[\sum_{i=1}^n \otimes c_{ij} \right]_{1 \times n} = [\otimes c_i]_{n \times 1}^T \quad (20)$$

Note: “superscript T” is the matrix transposed $\otimes r_i = (r_i^d, r_i^g, r_i^t)$ and $\otimes c_i = (c_i^d, c_i^g, c_i^t)$.

The index quantifying the strength of influences conveyed and received is $\otimes r_i + \otimes c_i$. The result of $\otimes r_i - \otimes c_i$ represents the net influence. A higher $\otimes r_i + \otimes c_i$ value indicates a higher degree of influence of factor i on the evaluation system. If $\otimes r_i - \otimes c_i > 0$ (is positive), then it is indicated that indicator i significantly influences others. If $\otimes r_i - \otimes c_i < 0$ (is negative), then other indicators have an impact on indicator i .

To de-fuzzy the fuzzy value ($\otimes \beta = (\beta^d + \beta^g + \beta^t)$) to calculate the crisp value (λ), the centroid is calculated by using Equation (21):

$$\beta = \frac{(\beta^d + \beta^g + \beta^t)}{3} \quad (21)$$

The indicator’s overall effect on the assessment system is represented by $\otimes r_i + \otimes c_i$. Therefore, Equation (22) will be used to construct the impact weight of an indicator.

$$\mathcal{E}_i = \frac{(r_i + c_i)}{\sum_{i=1}^n (r_i + c_i)} \quad (22)$$

3.2.4. Z-fuzzy TOPSIS method

Developed by Yoon and Hwang in 1981, TOPSIS is an MCDM tool that has been evaluated by decision-makers at different levels.

This method is based on the idea that the best alternative will be the closest to the positive-ideal strategies (PIS) and also the farthest from the negative-ideal strategies (NIS) [25]. This means that an alternative is rated higher than an alternative with a lower similarity value if it is more similar to the best strategies. Using this method, we can base on the distance of the options to PIS and NIS, thereby finding out which is the best strategy, and ranking the goodness of each strategy. This paper combines Z-fuzzy theory with TOPSIS to capture the uncertainty of the real-world assessment setting [37,70]. The calculation of TOPSIS in detail is explained below.

Step 1. Set the symbol

Assume there are m strategies $P_i = \{P_1, P_2, \dots, P_m\}$ and n factors $I_j = \{I_1, I_2, \dots, I_m\}$, and the weight of the factors is defined as $\mathcal{E}_j = \{\mathcal{E}_1, \mathcal{E}_2, \dots, \mathcal{E}_m\}$. Each expert E_k ($k = 1, 2, \dots, K$) assesses the performance of the strategy P_i to factor I_j .

Step 2. Create the initial fuzzy decision matrix (FDM) $\otimes E$

Every project is evaluated by the decision-maker using the scales shown in Table 6. By utilizing the arithmetic mean to average the assessment scores of each decision-maker, the initial assessment FDM is produced, as shown in Equation (23):

$$\otimes E = [\otimes e_{ij}]_{m \times n} = \begin{bmatrix} \otimes e_{11} & \otimes e_{12} & \dots & \otimes e_{1j} & \dots & \otimes e_{1n} \\ \otimes e_{21} & \otimes e_{22} & \dots & \otimes e_{2j} & \dots & \otimes e_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes e_{i1} & \otimes e_{i2} & \dots & \otimes e_{ij} & \dots & \otimes e_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes e_{m1} & \otimes e_{m2} & \dots & \otimes e_{mj} & \dots & \otimes e_{mn} \end{bmatrix} \quad (23)$$

Here $\otimes e_{ij} = (e_{ij}^d, e_{ij}^g, e_{ij}^t)$, where $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$; and $e_{ij}^d = \frac{1}{K} \sum_{k=1}^K e_{ij}^d$, $e_{ij}^g = \frac{1}{K} \sum_{k=1}^K e_{ij}^g$ and $e_{ij}^t = \frac{1}{K} \sum_{k=1}^K e_{ij}^t$ where $k = 1, 2, \dots, K$

Step 3. Calculate a normalized FDM $\otimes \widetilde{O}^*$

The normalized fuzzy matrix is $\otimes E^* = [e_{ij}^*]_{m \times n}$. Equation (24) illustrates how the conventional normalized method uses the project's best performance score as the denominator. The values in the matrix after normalization will range from 0 to 1.

$$e_{ij}^* = \frac{\otimes e_{ij}}{\max \{ \otimes e_{ij} \}} \quad (24)$$

Step 4. Compute a weighted formalized FDM $\otimes \widetilde{O}^{**}$

The weighted normalized FDM is produced by multiplying the weight ($\mathcal{E}_j$) determined by the indicator by the normalized FDM $\otimes \widetilde{O}^*$, taking into account the varying importance of each indicator. The calculation method is shown in Equation (25):

$$\otimes E^{**} = [\otimes e_{ij}^{**}]_{m \times n} = e_{ij}^* \cdot \mathcal{E}_j \quad (25)$$

Step 5. Calculate the fuzzy negative ideal strategy (FNIS) and fuzzy positive ideal strategy (FPIS) using Equation (26)–(29):

$$FPIS = E^{best} = (E_1^{best}, E_2^{best}, \dots, E_n^{best}) \quad (26)$$

$$FNIS = E^{worst} = (E_1^{worst}, E_2^{worst}, \dots, E_n^{worst}) \quad (27)$$

Where

$$E_j^{best} = (\max \{ e_{ij}^d \}, \max \{ e_{ij}^g \}, \max \{ e_{ij}^t \}) \quad (28)$$

$$E_j^{worst} = (\min \{ e_{ij}^d \}, \min \{ e_{ij}^g \}, \min \{ e_{ij}^t \}) \quad (29)$$

Step 6. Calculate the distance between each project strategy to the FPIS and FNIS.

Using Equations (30) and (31), the separation gaps between strategies FPIS, and FNIS are computed. This step involved defuzzifying the fuzzy scores and converting them to crisp scores.

$$\lambda_i = \sum_{j=1}^n \sqrt{\left(\frac{(E_j^{best} - e_{ij}^{*sd})^2 + 2 * (E_j^{best} - e_{ij}^{*sg})^2 + (E_j^{best} - e_{ij}^{*st})^2}{4} \right)} \quad (30)$$

$$\lambda_i^- = \sum_{j=1}^n \sqrt{\left(\frac{(e_{ij}^{*+d} - E_j^{worst})^2 + 2 * (e_{ij}^{*+g} - E_j^{worst})^2 + (e_{ij}^{*+t} - E_j^{worst})^2}{4} \right)} \quad (31)$$

Step 7. Calculate the closeness coefficient index (CC_i).

The closeness coefficient, CC_i , is a trustworthy ranking metric that shows the optimal degree of strategy. The greater this coefficient is, the more idea-driven the corresponding tactics are as they are closer to FPIS and farther from FNIS [37]. The CC_i ranking index solves the drawbacks of the traditional TOPSIS ranking index by accounting for the variations between all strategies and the FPIS and FNIS. Equation (32) can be used to determine the proximity coefficient.

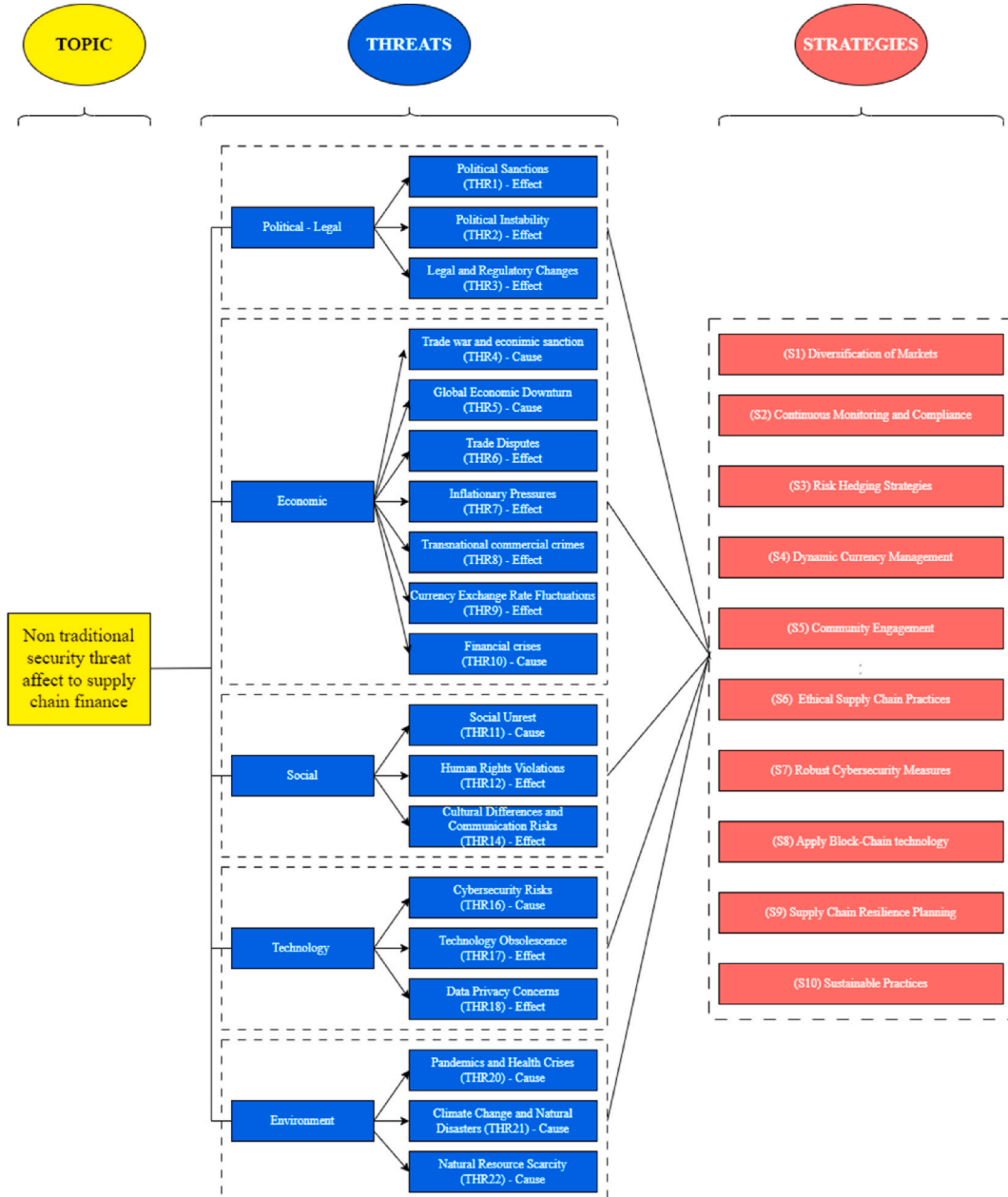


Fig. 2. Hierarchy of non-traditional security threats.

$$CC_i = \eta^+ \left(\frac{\lambda_i^-}{\sum_{i=1}^m \lambda_i^-} \right) - \eta^- \left(\frac{\lambda_i^+}{\sum_{i=1}^m \lambda_i^+} \right), \begin{cases} -1 \leq CC_i \leq 1 \\ 0 \leq \eta^+ \leq 1 \\ 0 \leq \eta^- \leq 1 \end{cases}, i = 1, 2, \dots, m \quad (32)$$

4. Case study

4.1. Research background

Non-traditional security is a challenge facing the whole world, including Vietnam. The dangers, obstacles, and threats that currently face security are becoming more obvious and have an immediate effect on every aspect of life, including defense, security, society, and the economy [74]. A typical example is the COVID-19 epidemic. According to a report by Vietnam General Statistics Office [1], in the first quarter of 2022, 16.9 million people were negatively affected by the pandemic, of which 0.9 million people lost their lives. Jobs, accounting for 1.2 % and 13.7 million workers had reduced income, accounting for 18.3 % thereby seriously affecting the situation of supply chains in Vietnam. The COVID-19 epidemic had a significant negative impact on Vietnam's economy in 2021. Particularly in the first nine months, regions such as the Southeast provinces, the Mekong Delta, and Hanoi experienced prolonged social distancing measures due to the epidemic, leading to a notable decline in GDP growth. The third quarter of 2021 witnessed the most substantial decrease in GDP since quarterly GDP calculations began in Vietnam, with a 6.17 % drop compared to the same period last year. Despite some sectors such as agriculture, forestry, and fisheries experiencing positive growth of 1.04 %, it remains notably low compared to the past decade. Thus, the COVID-19 epidemic has a profound impact on the entire economy as well as the operations of Vietnamese businesses.

Beyond epidemics, a spectrum of non-traditional security challenges, including natural disasters, cyber-attacks, trade wars, and transnational crimes ..., is currently exerting significant influence on Vietnam's financial supply chain [13]. These factors intricately impact production, business operations, human resources, and overall goods consumption, leading to disruptions in the financial supply chain's seamlessness and efficiency. A compromised financial supply chain efficiency, as emphasized by Beka Be Nguema et al. [6], culminates in a broader decline in the efficiency of the entire supply chain, thereby posing substantial adversities to the Vietnamese economy. Recognizing the imperative of addressing these challenges, this study aims to identify, measure, and propose response strategies. Based on the comprehensive data analysis from the three distinct phases, coupled with the categorization of validated factors through PESTEL analysis into distinct threat groups encompassing Political-Legal, Economic, Social, Technology, and Environmental aspects, and the subsequent ranking of proposed strategies as presented in Fig. 2.

4.2. Expert selection

The research was conducted within the territory of Vietnam, and the issues were raised in the Vietnamese context, and experts were not selected randomly but based on their knowledge and experience related to the topic. Previous research indicates that at least 10 to 18 experts should participate in the assessment for each area of expertise (P. H. [12]; P.-H. [30,72]). The study was conducted from December 2023 to February 2024. The survey was created on the Jotform platform and sent online to a team of experts. There were 54 survey invitations sent out via email and 30 responses received, reaching 56 %. Experts have at least 10 years of experience, including many lecturers and researchers at many famous universities, along with many leaders at famous corporations. All experts are over 25 years old and 30 % of experts have over 20 years of experience. Experts all have bachelor's degrees or higher, of which 50 % are masters and 43 % are doctors. The experts are all lecturers, researchers, and employees directly working in the field of supply chain management and logistics. Based on the above information, these experts completely satisfy the knowledge, experience, and ability requirements to participate in surveys in this field as shown in Table 9.

Table 9
Experts' profiles.

Age	From 25 to 40	2 persons	7 %
	From 40 to 60	19 persons	63 %
	Over 60	6 persons	20 %
Gender	Male	14 persons	47 %
	Female	13 persons	43 %
	Other	3 persons	10 %
Education	Bachelor	2 persons	7 %
	Master	15 persons	50 %
	Doctor	13 persons	43 %
Work	Teaching	10 persons	33 %
	Research	10 persons	33 %
	Working	10 persons	33 %
Experience	From 10 to 20 years	21 persons	70 %
	Over 20 years	9 persons	30 %

4.3. Results of Z-Fuzzy Delphi phase

In this study, the non-traditional security threats found through the literature review are 22 threats. These factors will be evaluated by experts for their suitability using linguistic scales. The survey is conducted online, part 1 will collect demographic information such as age, qualifications, experience, and job. The expert's answer will be converted from a linguistic variable to a Z-Fuzzy number according to Table 6. The expert evaluation results for each factor will be synthesized, and indexes will be calculated as presented in section 3. Based on comparing the V_c value of each threat with the threshold value, three factors, THR13, THR15, and THR19, were rejected. The remaining factors will be used in Phase 2. Table 10 presents the Z-Fuzzy Delphi result.

4.4. Result of Z-Fuzzy Dematel phase

Threats after being validated and accepted at the Z-Fuzzy Dematel phase will be analyzed for the relationships between them. The thirty experts above will participate in phase 2: evaluating the impact relationships of factors in pairs. The evaluation results of 30 experts will be synthesized using the average method, standardized, and create a total influence matrix as described in section 3. Table 11 presents the total influence matrix between factors.

Table 12 below shows the values of $\otimes r_i$, $\otimes c_i$, $\otimes r_i + \otimes c_i$ and $\otimes r_i - \otimes c_i$ as well as the weights $\mathcal{E}$ of each threat after defuzzy.

The main results indicate that threats THR4, THR5, THR10, THR11, THR16, THR20, THR21, and THR22 emerge as “cause” factors, signifying that their impact levels surpass the degree of influence, because their corresponding $\otimes r_i - \otimes c_i$ values are greater than zero. The value $\otimes r_i$ denotes the extent to which threat i affects other threats, while $\otimes c_i$ represents the degree to which threat i is influenced by other threats. A positive net impact value $\otimes r_i - \otimes c_i$ indicates that the influence of threat i outweighs its susceptibility to influence, signifying its role as a driving factor influencing or leading other threats. Conversely, threats THR1, THR2, THR3, THR6, THR7, THR8, THR9, THR12, THR14, THR17, and THR18 belong to the “effect” factor group, given their $\otimes r_i - \otimes c_i$ values below zero. Threats categorized in the effect group exhibit $\otimes r_i$ values smaller than $\otimes c_i$, indicating that their susceptibility to influence from other threats outweighs their capacity to influence other threats. Thus, they are regarded as factors that are influenced, impacted, or deemed consequences of other factors. Hence, discerning the causality of threats aids in evaluating their impact on SCF. Specifically, factors categorized in the cause group not only directly affect themselves but also precipitate or exacerbate numerous other threats, resulting in intricate ramifications. Conversely, effect factors manifest or intensify under the influence of cause factors, offering policymakers guidance on addressing root causes to mitigate associated challenges. In terms of influence weight, THR5 takes the lead as the most impactful factor, succeeded by THR20, THR10, and THR16, respectively. These are the four NTS threats that are assessed to have the greatest impact on SCF in Vietnam today. Conversely, threats THR22 and THR14 exhibit the lowest impact levels within the analyzed factors. By integrating influence weight and the classification of factors as “cause” or “effect,” we can pinpoint threats with substantial influence weight within the cause group as root causes. These threats not only exert a significant impact on SCF but also precipitate or exacerbate numerous other threats, exhibiting a dual effect. Therefore, stakeholders should prioritize addressing these root causes to effectively mitigate associated challenges.

Fig. 3 illustrates the intricate relationships among the top 10 factors, showcasing varying degrees of influence. The chart categorizes these influences into three tiers: strong, medium, and weak. The determination of influence levels relies on the max, higher,

Table 10
The Z-Fuzzy Delphi method results.

Factor	Aggregate	u_c	l_c	V_c	Validate
THR1	(0.0316, 0.2055, 0.3795)	0.2925	−0.0554	0.1324	Accepted
THR2	(0, 0.182, 0.3795)	0.2808	−0.091	0.1177	Accepted
THR3	(0, 0.1698, 0.3795)	0.2747	−0.0849	0.1161	Accepted
THR4	(0, 0.1598, 0.3795)	0.2697	−0.0799	0.1149	Accepted
THR5	(0, 0.2057, 0.3795)	0.2926	−0.1029	0.1206	Accepted
THR6	(0, 0.1664, 0.3795)	0.273	−0.0832	0.1157	Accepted
THR7	(0, 0.1877, 0.3795)	0.2836	−0.0939	0.1183	Accepted
THR8	(0, 0.1676, 0.3795)	0.2736	−0.0838	0.1159	Accepted
THR9	(0, 0.195, 0.3795)	0.2873	−0.0975	0.1193	Accepted
THR10	(0, 0.222, 0.3795)	0.3008	−0.111	0.1227	Accepted
THR11	(0, 0.1604, 0.3795)	0.27	−0.0802	0.115	Accepted
THR12	(0, 0.1905, 0.3795)	0.285	−0.0953	0.1187	Accepted
THR13	(0, 0, 0.2121)	0.1061	0	0.0531	Rejected
THR14	(0, 0.2055, 0.3795)	0.2925	−0.1028	0.1206	Accepted
THR15	(0, 0, 0.2121)	0.1061	0	0.0531	Rejected
THR16	(0, 0.1659, 0.3795)	0.2727	−0.083	0.1156	Accepted
THR17	(0, 0.1711, 0.3795)	0.2753	−0.0856	0.1163	Accepted
THR18	(0, 0.1421, 0.3795)	0.2608	−0.0711	0.1126	Accepted
THR19	(0, 0, 0.1897)	0.0949	0	0.0475	Rejected
THR20	(0, 0.1963, 0.3795)	0.2879	−0.0982	0.1194	Accepted
THR21	(0, 0.1996, 0.3795)	0.2896	−0.0998	0.1199	Accepted
THR22	(0.0548, 0.1917, 0.3795)	0.2856	−0.0137	0.1394	Accepted
Threshold				0.1102	

Table 11

Total influence matrix after defuzzy.

	THR1	THR2	THR3	THR4	THR5	THR6	THR7	THR8	THR9	THR10	THR11	THR12	THR14	THR16	THR17	THR18	THR20	THR21	THR22
THR1	0.1437	0.1924	0.175	0.1608	0.1651	0.1832	0.1816	0.1829	0.1919	0.1688	0.1634	0.1801	0.1661	0.1728	0.1808	0.1799	0.1646	0.1354	0.1364
THR2	0.188	0.165	0.1875	0.1685	0.178	0.1921	0.1967	0.197	0.2007	0.1825	0.1777	0.2024	0.1788	0.1783	0.1856	0.2017	0.1691	0.1413	0.1436
THR3	0.1756	0.1941	0.1488	0.1616	0.1722	0.1949	0.1865	0.1905	0.1904	0.1757	0.1698	0.1866	0.1739	0.1723	0.182	0.1855	0.1706	0.1482	0.1462
THR4	0.2302	0.2413	0.2176	0.1615	0.2151	0.2351	0.2386	0.2333	0.2385	0.217	0.2091	0.2322	0.2152	0.2196	0.2233	0.2197	0.2138	0.1768	0.1708
THR5	0.2167	0.2389	0.2211	0.1911	0.1695	0.23	0.231	0.2308	0.2359	0.2144	0.2027	0.2161	0.2029	0.2135	0.2187	0.2204	0.206	0.1786	0.1676
THR6	0.192	0.196	0.186	0.1704	0.1802	0.1609	0.1938	0.2009	0.2003	0.1791	0.1707	0.1962	0.1805	0.1798	0.1913	0.1892	0.1917	0.1508	0.143
THR7	0.1935	0.2084	0.1911	0.1685	0.1868	0.2024	0.162	0.198	0.2125	0.1839	0.1805	0.1989	0.1831	0.177	0.1921	0.1949	0.1889	0.152	0.1435
THR8	0.1837	0.1988	0.1893	0.1673	0.1706	0.1988	0.1885	0.1587	0.1987	0.1758	0.1747	0.1921	0.182	0.1823	0.1883	0.1894	0.1864	0.1492	0.1401
THR9	0.1933	0.2035	0.1952	0.1682	0.1861	0.2012	0.208	0.1989	0.1659	0.1901	0.1795	0.1988	0.179	0.1754	0.1886	0.1878	0.1872	0.1532	0.1456
THR10	0.2175	0.2321	0.2113	0.1893	0.2112	0.2232	0.2227	0.2212	0.229	0.1676	0.2004	0.2218	0.2008	0.2065	0.2119	0.2126	0.2051	0.1688	0.1579
THR11	0.2035	0.2209	0.198	0.18	0.1917	0.2065	0.2067	0.2083	0.2112	0.1968	0.1545	0.218	0.1957	0.1984	0.2056	0.2039	0.1954	0.1607	0.1585
THR12	0.1802	0.1931	0.1818	0.1576	0.1608	0.1838	0.1829	0.1842	0.1823	0.1682	0.1661	0.1499	0.1732	0.1672	0.1801	0.1764	0.1801	0.1455	0.1321
THR14	0.1716	0.1931	0.1787	0.1571	0.1632	0.1905	0.1764	0.1808	0.1873	0.1682	0.166	0.1886	0.1395	0.1726	0.177	0.1859	0.1767	0.1515	0.1399
THR16	0.213	0.2266	0.2123	0.1891	0.2029	0.213	0.2175	0.223	0.216	0.2081	0.1972	0.2099	0.2048	0.1659	0.2204	0.218	0.2064	0.1707	0.1622
THR17	0.1768	0.1908	0.1776	0.1573	0.1721	0.1966	0.1844	0.1908	0.1942	0.1703	0.1624	0.1906	0.1702	0.1895	0.15	0.1925	0.1805	0.1476	0.1407
THR18	0.1877	0.2042	0.1899	0.164	0.1701	0.2007	0.1919	0.1979	0.1956	0.1778	0.1679	0.1932	0.1728	0.1858	0.1928	0.1538	0.1838	0.1491	0.1393
THR20	0.2135	0.2317	0.2096	0.1913	0.2097	0.216	0.2223	0.2232	0.229	0.2104	0.2053	0.2207	0.2071	0.2075	0.2104	0.2067	0.1694	0.1812	0.1615
THR21	0.2109	0.2347	0.2144	0.1859	0.2095	0.2291	0.221	0.2222	0.2319	0.2123	0.2069	0.2166	0.1992	0.202	0.2057	0.2068	0.2139	0.1417	0.1733
THR22	0.2088	0.2323	0.2096	0.1884	0.2073	0.2227	0.217	0.222	0.2224	0.2087	0.198	0.2217	0.1978	0.2049	0.2084	0.2106	0.204	0.1761	0.1336

Table 12
Z-Fuzzy Dematel results.

Threats	$\otimes r_i$	$\otimes c_i$	$\otimes r_i + \otimes c_i$	Weight	Rank	$\otimes r_i - \otimes c_i$	Relations
THR1	3.2252	3.7002	6.9254	0.05054	16	-0.475	Effect
THR2	3.4343	3.9979	7.4322	0.05424	6	-0.5636	Effect
THR3	3.3255	3.6947	7.0202	0.05123	15	-0.3692	Effect
THR4	4.1087	3.2781	7.3868	0.05391	7	0.8306	Cause
THR5	4.006	3.5223	7.5283	0.05494	1	0.4837	Cause
THR6	3.453	3.8806	7.3336	0.05352	9	-0.4276	Effect
THR7	3.5182	3.8293	7.3475	0.05362	8	-0.3111	Effect
THR8	3.4148	3.8648	7.2796	0.05312	10	-0.45	Effect
THR9	3.5056	3.9337	7.4393	0.05429	5	-0.4281	Effect
THR10	3.9109	3.5758	7.4867	0.05464	3	0.3351	Cause
THR11	3.7142	3.4528	7.167	0.0523	11	0.2614	Cause
THR12	3.2456	3.8345	7.0801	0.05167	13	-0.5889	Effect
THR14	3.2646	3.5226	6.7872	0.04953	18	-0.258	Effect
THR16	3.877	3.5713	7.4483	0.05435	4	0.3057	Cause
THR17	3.3351	3.7132	7.0483	0.05144	14	-0.3781	Effect
THR18	3.4183	3.7358	7.1541	0.05221	12	-0.3175	Effect
THR20	3.9263	3.5938	7.5201	0.05488	2	0.3325	Cause
THR21	3.9377	2.9784	6.9161	0.05047	17	0.9593	Cause
THR22	3.8943	2.8356	6.7299	0.04911	19	1.0587	Cause

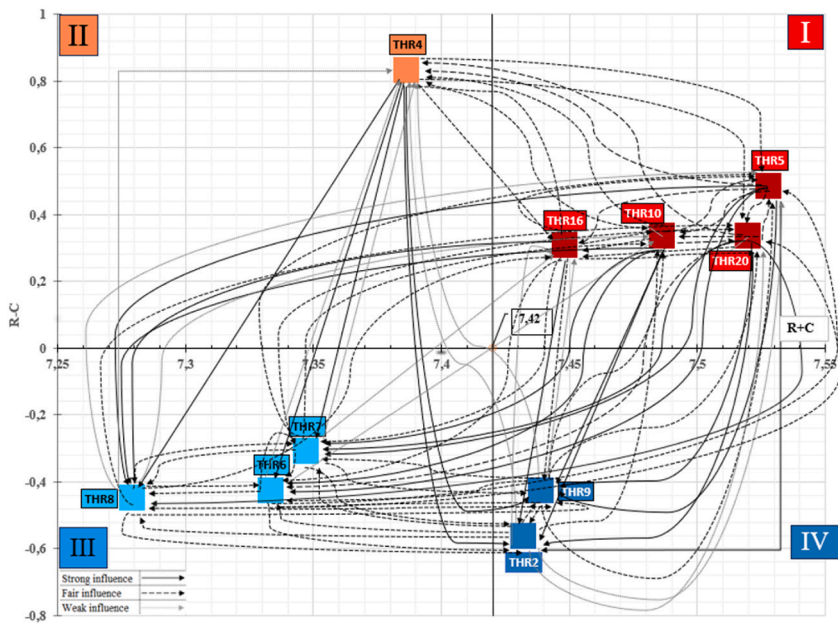


Fig. 3. INRM of top 10 threats.

lower, and min values derived from the comprehensive INRM. Specifically, the values of 0.2413 (max), 0.2207 (higher), 0.1794 (lower), and 0.1587 (min) serve as thresholds. In this context, influence values spanning from min to lower signify a low-level relationship, those spanning from lower to higher denote a medium-level relationship, while the remainder signifies high-level relationships. Factors in the cause group are highlighted in red, while those in the effect group are shown in blue. Threat groups with stronger impacts are indicated with darker colors.

The average $\otimes r_i + \otimes c_i$ level value of the 10 factors is used as the vertical axis to divide the data into two groups: strong influence and weak influence. Thus, the factors in the first quadrant including THR5 - global economic downturn, THR10 - financial crises, THR20 - pandemic and health crises, and THR16 - cybersecurity risk are all cause factors and have a strong influence, so these can be identified as root causes. Thus, three causes belong to the group of macro-financial risks, in addition, cybersecurity issues and epidemics are also assessed as root causes affecting the financial supply chain in Vietnam. THR4-trade war and economic sanction located in the second quadrant is also a factor in the cause group, but the level of influence is not as high as the group of four factors mentioned above. THR9 - currency exchange rate fluctuation and THR2-political instability are located in the 4th quadrant of the effect group but have a high level of importance and influence, while THR6 - trade dispute, THR7 - inflation pressures, THR8-transactional commercial crimes are also effects with a level of influence not as high as the two factors mentioned above.

Table 13
Weighted normalized matrix and FPIS, FNIS.

	THR1	THR2	THR3	THR4	THR5	THR6	THR7	THR8	THR9	THR10	THR11	THR12	THR14	THR16	THR17	THR18	THR20	THR21	THR22
S1	(0.0252, 0.0304, 0.0356)	(0.0219, 0.0277, 0.0336)	(0.0249, 0.0304, 0.0359)	(0.0377, 0.0441, 0.0506)	(0.0355, 0.0417, 0.0479)	(0.0269, 0.0327, 0.0385)	(0.0216, 0.0274, 0.0332)	(0.021, 0.0268, 0.0327)	(0.0296, 0.0358, 0.042)	(0.0318, 0.0378, 0.0437)	(0.0219, 0.0274, 0.033)	(0.0261, 0.032, 0.0378)	(0.0295, 0.0354, 0.0412)	(0.0255, 0.0317, 0.0378)	(0.0268, 0.0326, 0.0384)	(0.0217, 0.0275, 0.0333)	(0.021, 0.0271, 0.0333)	(0.0256, 0.0313, 0.037)	(0.0298, 0.0354, 0.041)
S2	(0.0292, 0.0349, 0.0406)	(0.0319, 0.0378, 0.0437)	(0.0361, 0.0418, 0.0474)	(0.029, 0.0351, 0.0412)	(0.0324, 0.0385, 0.0446)	(0.0321, 0.038, 0.0438)	(0.0261, 0.0319, 0.0377)	(0.0345, 0.0403, 0.0461)	(0.0313, 0.0374, 0.0435)	(0.0353, 0.0417, 0.048)	(0.038, 0.0439, 0.0497)	(0.0306, 0.0364, 0.0422)	(0.0277, 0.0332, 0.0388)	(0.0365, 0.0428, 0.0491)	(0.0291, 0.0351, 0.041)	(0.0305, 0.0364, 0.0424)	(0.0351, 0.0414, 0.0476)	(0.0279, 0.0337, 0.0396)	(0.0338, 0.0395, 0.0452)
S3	(0.0315, 0.0369, 0.0424)	(0.0375, 0.0436, 0.0498)	(0.0245, 0.0303, 0.0361)	(0.0362, 0.0423, 0.0483)	(0.0402, 0.0464, 0.0527)	(0.032, 0.0379, 0.0438)	(0.0359, 0.0424, 0.0488)	(0.0259, 0.0318, 0.0377)	(0.0367, 0.043, 0.0493)	(0.0375, 0.0436, 0.0498)	(0.0331, 0.0389, 0.0447)	(0.031, 0.0366, 0.0421)	(0.0268, 0.0321, 0.0374)	(0.0364, 0.0424, 0.0483)	(0.0272, 0.0326, 0.0381)	(0.0329, 0.0386, 0.0443)	(0.0339, 0.0402, 0.0465)	(0.0326, 0.0383, 0.0439)	(0.0325, 0.0383, 0.044)
S4	(0.0233, 0.0289, 0.0345)	(0.0273, 0.0331, 0.0389)	(0.028, 0.0337, 0.0394)	(0.0343, 0.0403, 0.0462)	(0.0362, 0.0425, 0.0487)	(0.0276, 0.0335, 0.0395)	(0.034, 0.0403, 0.0466)	(0.028, 0.0338, 0.0396)	(0.038, 0.0444, 0.0508)	(0.0373, 0.0436, 0.05)	(0.0294, 0.0353, 0.0412)	(0.0246, 0.0303, 0.036)	(0.0249, 0.0305, 0.0361)	(0.0292, 0.0353, 0.0415)	(0.0266, 0.0324, 0.0383)	(0.0233, 0.0292, 0.035)	(0.0291, 0.0356, 0.0422)	(0.0232, 0.0291, 0.0349)	(0.0269, 0.0327, 0.0385)
S5	(0.0258, 0.0315, 0.0371)	(0.0372, 0.0433, 0.0494)	(0.0269, 0.0329, 0.0389)	(0.0315, 0.0378, 0.044)	(0.0294, 0.0357, 0.0419)	(0.0318, 0.0379, 0.0439)	(0.0285, 0.0346, 0.0407)	(0.0325, 0.0383, 0.045)	(0.0288, 0.0349, 0.0411)	(0.0296, 0.0355, 0.0415)	(0.0366, 0.0425, 0.0484)	(0.0327, 0.0383, 0.044)	(0.0327, 0.0383, 0.044)	(0.025, 0.0305, 0.0361)	(0.0276, 0.0335, 0.0394)	(0.0312, 0.0367, 0.0423)	(0.0255, 0.0312, 0.0368)	(0.0306, 0.0366, 0.0427)	(0.031, 0.0368, 0.0426)
S6	(0.0294, 0.0351, 0.0409)	(0.0282, 0.0344, 0.0407)	(0.0289, 0.0349, 0.0409)	(0.033, 0.039, 0.0451)	(0.0304, 0.0368, 0.0432)	(0.0317, 0.038, 0.0444)	(0.0266, 0.0328, 0.0389)	(0.0341, 0.0402, 0.0463)	(0.0267, 0.0328, 0.039)	(0.0311, 0.037, 0.043)	(0.0223, 0.0282, 0.0341)	(0.0216, 0.0277, 0.0338)	(0.0192, 0.0247, 0.0302)	(0.0301, 0.0364, 0.0427)	(0.0227, 0.0287, 0.0346)	(0.028, 0.0339, 0.0398)	(0.0224, 0.0284, 0.0344)	(0.0296, 0.0354, 0.0411)	(0.027, 0.0326, 0.0381)
S7	(0.0261, 0.0318, 0.0375)	(0.0324, 0.0387, 0.0449)	(0.0315, 0.0375, 0.0435)	(0.0264, 0.0323, 0.0382)	(0.0321, 0.0384, 0.0448)	(0.0319, 0.038, 0.0441)	(0.0264, 0.0327, 0.039)	(0.0364, 0.0426, 0.0487)	(0.0289, 0.0349, 0.0408)	(0.0293, 0.0355, 0.0417)	(0.0292, 0.0349, 0.0406)	(0.0281, 0.034, 0.04)	(0.025, 0.0303, 0.0355)	(0.0355, 0.0419, 0.0482)	(0.0309, 0.0369, 0.043)	(0.0322, 0.0381, 0.0439)	(0.026, 0.032, 0.038)	(0.0226, 0.0284, 0.0342)	(0.0254, 0.0309, 0.0364)
S8	(0.0252, 0.0313, 0.0374)	(0.0307, 0.0368, 0.0429)	(0.028, 0.0337, 0.0394)	(0.0322, 0.0382, 0.0442)	(0.028, 0.0337, 0.0395)	(0.0272, 0.033, 0.0388)	(0.027, 0.0325, 0.0381)	(0.0304, 0.0365, 0.0425)	(0.0276, 0.0338, 0.0399)	(0.0327, 0.0388, 0.0449)	(0.0272, 0.0333, 0.0393)	(0.0244, 0.0302, 0.036)	(0.0241, 0.0296, 0.0352)	(0.0354, 0.0415, 0.0477)	(0.0306, 0.0365, 0.0424)	(0.0302, 0.0361, 0.042)	(0.032, 0.0383, 0.0445)	(0.0276, 0.0332, 0.0388)	(0.0301, 0.0359, 0.0417)
S9	(0.027, 0.0328, 0.0386)	(0.0298, 0.036, 0.0422)	(0.0333, 0.0393, 0.0453)	(0.0327, 0.0391, 0.0456)	(0.0393, 0.0457, 0.052)	(0.0312, 0.0376, 0.0439)	(0.0297, 0.0358, 0.0419)	(0.028, 0.0344, 0.0408)	(0.0325, 0.0387, 0.0448)	(0.0336, 0.0399, 0.0462)	(0.0245, 0.0303, 0.0361)	(0.0243, 0.0303, 0.0363)	(0.0233, 0.029, 0.0348)	(0.0278, 0.0342, 0.0405)	(0.0242, 0.0302, 0.0361)	(0.0275, 0.0333, 0.0392)	(0.0294, 0.0359, 0.0423)	(0.0312, 0.0373, 0.0433)	(0.0289, 0.0344, 0.04)
S10	(0.0341, 0.0402, 0.0463)	(0.0366, 0.043, 0.0494)	(0.0288, 0.0346, 0.0404)	(0.0331, 0.0393, 0.0455)	(0.0365, 0.0429, 0.0493)	(0.0293, 0.0353, 0.0414)	(0.0267, 0.0328, 0.0389)	(0.034, 0.0402, 0.0464)	(0.0297, 0.0356, 0.0422)	(0.0361, 0.0425, 0.0488)	(0.0297, 0.0356, 0.0415)	(0.0339, 0.0398, 0.0458)	(0.0251, 0.0307, 0.0364)	(0.0373, 0.0437, 0.0501)	(0.0321, 0.038, 0.044)	(0.0324, 0.0385, 0.0446)	(0.0325, 0.0389, 0.0453)	(0.0333, 0.0391, 0.0448)	(0.0374, 0.0433, 0.0491)
FPIS	(0.0341, 0.0402, 0.0463)	(0.0375, 0.0436, 0.0498)	(0.0361, 0.0418, 0.0474)	(0.0377, 0.0441, 0.0506)	(0.0402, 0.0464, 0.0527)	(0.0321, 0.038, 0.0444)	(0.0359, 0.0424, 0.0488)	(0.0364, 0.0426, 0.0487)	(0.038, 0.0444, 0.0508)	(0.0375, 0.0436, 0.05)	(0.038, 0.0439, 0.0497)	(0.0339, 0.0398, 0.0458)	(0.0295, 0.0354, 0.0412)	(0.0373, 0.0437, 0.0501)	(0.0321, 0.038, 0.044)	(0.0329, 0.0386, 0.0446)	(0.0351, 0.0414, 0.0476)	(0.0333, 0.0391, 0.0448)	(0.0374, 0.0433, 0.0491)
FNIS	(0.0233, 0.0289, 0.0345)	(0.0219, 0.0277, 0.0336)	(0.0245, 0.0303, 0.0359)	(0.0264, 0.0323, 0.0382)	(0.028, 0.0337, 0.0395)	(0.0269, 0.0327, 0.0385)	(0.0216, 0.0274, 0.0332)	(0.021, 0.0268, 0.0327)	(0.0296, 0.0358, 0.042)	(0.0318, 0.0378, 0.0437)	(0.0219, 0.0274, 0.033)	(0.0261, 0.032, 0.0378)	(0.0295, 0.0354, 0.0412)	(0.0255, 0.0317, 0.0378)	(0.0268, 0.0326, 0.0384)	(0.0217, 0.0275, 0.0333)	(0.021, 0.0271, 0.0333)	(0.0256, 0.0313, 0.037)	(0.0298, 0.0354, 0.041)

4.5. Result of Z-fuzzy TOPSIS

Proposed strategies will be evaluated by experts for their suitability in addressing the negative impacts caused by non-traditional security challenges. A set of criteria including 19 threats from phase 2 will be used for evaluation. A group of 30 experts will evaluate how each strategy impacts each threat. The results are then synthesized using the arithmetic average method and calculating the steps according to the TOPSIS method as presented in section 3. Weighted normalized matrix and FPIS, FNIS are presented in Table 13 below.

The CC_i value results of each strategy and the ranking of strategies are presented in Table 14 below:

Table 14 outlines the ranking of proposed strategies based on their effectiveness in addressing non-traditional security challenges within the financial supply chain. The highest ranked strategies emerge as S10 – sustainable practice. Following closely is S3 – risk hedging strategy, securing the second position, while the third-ranking strategy is S2 – continuous monitoring and compliance. These top three strategies are deemed most adept in responding to the complexities posed by non-traditional security threats. Conversely, S1 – diversification of the market is identified as the least effective strategy, with S6 trailing closely behind in efficacy. The following is a summary of the strategies' order of priority: $S10 > S3 > S2 > S5 > S9 > S7 > S8 > S4 > S6 > S1$.

4.6. Comparative analysis

Comparative analysis was conducted to check the accuracy of the results ranked using the TOPSIS method. The results calculated using the Fuzzy TOPSIS, TOPSIS, and Z-fuzzy, Fuzzy, MOORA methods are used to compare with the ranking results of this study. Table 15 and Fig. 4 below present the results of six methods.

In the comprehensive analysis, all six methods consistently rank strategies S2, S3, S5, and S10 within the top four. While there may be minor discrepancies in the exact order of the top three, all methods unanimously highlight these four strategies as the most optimal choices for addressing nontraditional security threats and ensuring the effectiveness of supply chain finance. Strategies S1 and S6 consistently occupy positions outside the top 9 and top 10 in all three methods, indicating their comparatively lower suitability as strategies. Two MCDM approaches using various ranking algorithms are TOPSIS and MOORA. By using geometric distance, TOPSIS finds the best option that is the furthest from any negative-ideal solution and the closest to the positive-ideal solution. MOORA, on the other hand, uses a ratio-based approach, comparing the response of each alternative to a denominator that represents all alternatives for every objective. The discrepancy in ranking algorithms between these methods can lead to variations in results. Additionally, integrating the Z-Fuzzy concept enhances accuracy by considering the degree of certainty, ensuring more precise outcomes compared to approaches that overlook this aspect.

To analyze the correlation more accurately, Table 16 presents the results of the correlation tests between ranking orders obtained from different methods. Kendall's tau b and Spearman's rho correlation tests reveal that all correlation coefficient values surpass 0.7. Notably, the comparison of TOPSIS, fuzzy MOORA, and MOORA yields a correlation coefficient of 1, signifying identical ranking results from these three methods. The significance values for the remaining pairs of methods are all below 0.05, indicating a strong positive correlation in their ranking outcomes [75,76]. Similarly, the Pearson correlation result is all sig values are below 0.05, and the correlation values between pairs of methods exceed 0.8, demonstrating that the results of these ranking methods are related to each other [77]. These consistent findings across multiple tests affirm the reliability of the study's results despite minor variations in ranking outcomes among the methods.

4.7. Discussions

The main findings reveal the existence of 19 impactful NTS factors affecting the financial supply chains of businesses in Vietnam. These challenges span diverse political, legal, economic, technological, and environmental domains. Noteworthy strategies identified by experts, such as sustainable practices, risk hedging, and continuous monitoring with compliance, demonstrate efficacy in mitigating the damages inflicted by NTS risks on the financial supply chains of Vietnamese enterprises. This result is achieved using a combination of the Z number concept, fuzzy number, and MCDM methods: DELPHI, DEMATEL, and TOPSIS. Integrating the Z number with the Fuzzy number will help access information about the survey participant's level of certainty when answering that question, thereby

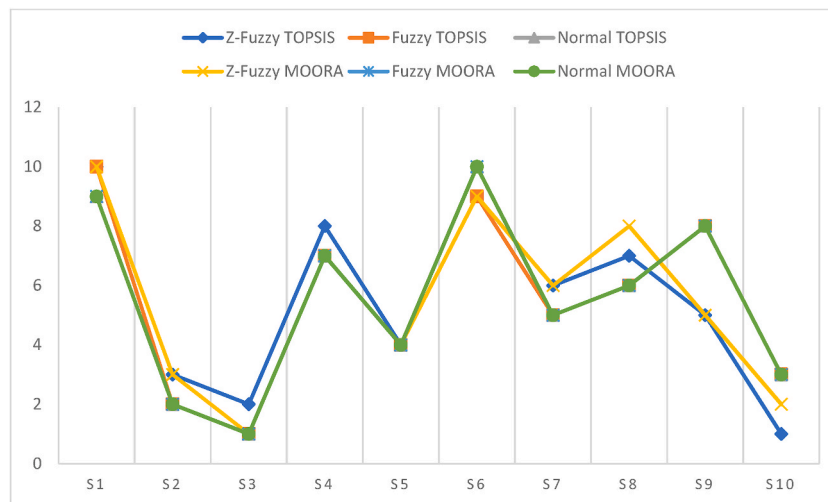
Table 14
The results of Z-Fuzzy TOPSIS.

Strategies	η^+	η^-	CC_i	Rank
S1	0.0635	0.1594	-0.0959	10
S2	0.1272	0.0655	0.0617	3
S3	0.1396	0.0725	0.0671	2
S4	0.0862	0.1195	-0.0333	8
S5	0.1004	0.091	0.0094	4
S6	0.076	0.1269	-0.0509	9
S7	0.0906	0.1023	-0.0117	6
S8	0.0848	0.1037	-0.0189	7
S9	0.0933	0.1008	-0.0075	5
S10	0.1383	0.0583	0.08	1

Table 15

Compare ranking using Z-Fuzzy, Fuzzy, TOPSIS and Z-Fuzzy, Fuzzy, MOORA.

STRATEGIES	Z-Fuzzy TOPSIS	Fuzzy TOPSIS	TOPSIS	Z-Fuzzy MOORA	Fuzzy MOORA	MOORA
S1	10	10	9	10	9	9
S2	3	2	2	3	2	2
S3	2	1	1	1	1	1
S4	8	7	7	7	7	7
S5	4	4	4	4	4	4
S6	9	9	10	9	10	10
S7	6	5	5	6	5	5
S8	7	6	6	8	6	6
S9	5	8	8	5	8	8
S10	1	3	3	2	3	3

**Fig. 4.** Compare between Z-Fuzzy, fuzzy, normal TOPSIS, and Z-Fuzzy, fuzzy, MOORA.

improving the accuracy of the data [37,46]. This combination is an improvement over previous studies such as Khan et al. [39], Kumar and Kumar Barua [7], Tseng et al. [43], and Wan Ahmad et al. [11], which only used fuzzy numbers or scrip scores. Using DELPHI to validate scales also increases the reliability of the factors used for the subsequent phases, which is an improvement compared to previous studies that did not validate factors such as Beka Be Nguema et al. [6], Kumar and Kumar Barua [7], Wang et al. [42]. At the same time, with the combination of TOPSIS and DEMATEL methods, the results show the weight of threats and find the cause-and-effect relationship between those factors. At the same time, the found weights serve as ranking input data for potential strategies. Compared to previous studies on the same topic, such as Akter et al. [40], Kumar and Kumar Barua [7], Mzougui et al. [9], and Tseng et al. [43] combining these two methods provides a more comprehensive picture of NTS threats affecting SCF and potential strategies.

The Z-Fuzzy Delphi findings indicate that migration and demographic change, transportation, and other logistic risks, as well as terrorism, exhibit minimal to no impact on Vietnam's Supply Chain Finance (SCF). Terrorism, recognized as a non-traditional problem, exerts negligible influence on the country's financial supply chain due to Vietnam's established political stability and robust security measures. The imperative lies in preserving Vietnam's stability and sustainable development to mitigate potential harm from terrorism (V. T. [12]). Therefore, the financial supply chain is not affected by the terrorism issue in the current Vietnamese context. Regarding migrations and demographic change, Vietnam is currently experiencing a demographic advantage with a substantial working-age population [78]. Migration and immigration are strictly managed, and this ensures Vietnam's human resources. In previous research, Giannakis and Papadopoulos [51] also pointed out that demographic challenges or the aging population have a low impact on the sustainability of the supply chain, which in turn has a low impact on SCF. Therefore, migrations and demographic change are not challenges affecting the financial supply chain in Vietnam. Examining transportation risks and other logistic risks, transportation is not prominently considered within the realm of non-traditional security. Its primary focus is on efficiently facilitating the movement of people and goods. Transportation security in Vietnam predominantly revolves around preventing traffic accidents and safeguarding infrastructure from potential threats. Consequently, transportation security addresses specific concerns within its domain, diverging from traditional security and non-traditional security.

According to the main finding of the Z-Fuzzy Dematel model, in the top 10, cause factors include THR5, THR20, THR10, THR16, and THR4, while effect factors comprise THR9, THR2, THR7, THR6, and THR8. THR5 – global economic downturn, classified as a

Table 16
Correlations analysis.

			Z-Fuzzy TOPSIS	Fuzzy TOPSIS	TOPSIS	Z-Fuzzy MOORA	Fuzzy MOORA	MOORA
Kendall's tau_b	Z-Fuzzy TOPSIS	Correlation	1.000	0.778**	0.733**	0.911**	0.733**	0.733**
		Sig. (2-tailed)		0.002	0.003	0.000	0.003	0.003
	Fuzzy TOPSIS	Correlation	0.778**	1.000	0.956**	0.778**	0.956**	0.956**
		Sig. (2-tailed)	0.002		0.000	0.002	0.000	0.000
	Normal TOPSIS	Correlation	0.733**	0.956**	1.000	0.733**	1.000**	1.000**
		Sig. (2-tailed)	0.003	0.000		0.003		
	Z-Fuzzy MOORA	Correlation	0.911**	0.778**	0.733**	1.000	0.733**	0.733**
		Sig. (2-tailed)	0.000	0.002	0.003		0.003	0.003
	Fuzzy MOORA	Correlation	0.733**	0.956**	1.000**	0.733**	1.000	1.000**
		Sig. (2-tailed)	0.003	0.000		0.003		
	Normal MOORA	Correlation	0.733**	0.956**	1.000**	0.733**	1.000**	1.000
		Sig. (2-tailed)	0.003	0.000		0.003		
Spearman's rho	Z-Fuzzy TOPSIS	Correlation	1.000	0.891**	0.879**	0.976**	0.879**	0.879**
		Sig. (2-tailed)		0.001	0.001	0.000	0.001	0.001
	Fuzzy TOPSIS	Correlation	0.891**	1.000	0.988**	0.903**	0.988**	0.988**
		Sig. (2-tailed)	0.001		0.000	0.000	0.000	0.000
	Normal TOPSIS	Correlation	0.879**	0.988**	1.000	0.891**	1.000**	1.000**
		Sig. (2-tailed)	0.001	0.000		0.001		
	Z-Fuzzy MOORA	Correlation	0.976**	0.903**	0.891**	1.000	0.891**	0.891**
		Sig. (2-tailed)	0.000	0.000	0.001		0.001	0.001
	Fuzzy MOORA	Correlation	0.879**	0.988**	1.000**	0.891**	1.000	1.000**
		Sig. (2-tailed)	0.001	0.000		0.001		
	Normal MOORA	Correlation	0.879**	0.988**	1.000**	0.891**	1.000**	1.000
		Sig. (2-tailed)	0.001	0.000		0.001		
Pearson Correlation	Z-Fuzzy TOPSIS	Correlation	1	0.891**	0.879**	0.976**	0.879**	0.879**
		Sig. (2-tailed)		0.001	0.001	0.000	0.001	0.001
	Fuzzy TOPSIS	Correlation	0.891**	1	0.988**	0.903**	0.988**	0.988**
		Sig. (2-tailed)	0.001		0.000	0.000	0.000	0.000
	Normal TOPSIS	Correlation	0.879**	0.988**	1	0.891**	1.000**	1.000**
		Sig. (2-tailed)	0.001	0.000		0.001	0.000	0.000
	Z-Fuzzy MOORA	Correlation	0.976**	0.903**	0.891**	1	0.891**	0.891**
		Sig. (2-tailed)	0.000	0.000	0.001		0.001	0.001
	Fuzzy MOORA	Correlation	0.879**	0.988**	1.000**	0.891**	1	1.000**
		Sig. (2-tailed)	0.001	0.000	0.000	0.001		0.000
	Normal MOORA	Correlation	0.879**	0.988**	1.000**	0.891**	1.000**	1
		Sig. (2-tailed)	0.001	0.000	0.000	0.001	0.000	

cause, has the greatest influence weight. This result aligns with Krause and Ellram [79], Wetzel and Hofmann [54], and Giannakis and Papadopoulos [51] have indicated that the downturn of the global economy has a large negative impact on global supply chains and supply chain finance. Global economic downturn is an inevitable challenge in the development of any nation, as underscored by recent global events after the Covid-19 pandemic. The downturn economic has significantly affected the global economy, especially in Europe, the United States, and China, profoundly impacting Vietnam's international financial supply chain (2022). THR20 - Pandemics and health crises are a global problem that strongly affects the supply chain financially. After the COVID-19 pandemic, this issue has received even more attention, and its effects have become extremely special in chain applications. This study found that pandemics and health crises as worthy of ranking 2nd in the top 10 risks with the most significant impact on the main financial supply chain. Similar to the results of this study, Giannakis and Papadopoulos [51] have also indicated that the pandemic has a great impact on the supply chain and financial supply chain. The research findings have also shown that THR10 - financial crisis is the third strongest influencing factor on the financial supply chain in Vietnam. Studies by Wetzel and Hofmann [54] and Giannakis and Papadopoulos [51] align with this result, emphasizing the centrality of financial crises in affecting supply chains and also on the efficiency of SCF. In Vietnam, the Minister of Industry and Trade issued a report demonstrating the impact of the financial crisis on the Vietnamese economy, stating that supply chain efficiency is deeply and strongly affected in 2020 (2021). Results in Table 12 also show that THR16-cybersecurity risks are ranked fourth and classified as a cause factor. This result is similar to previous research findings such as Stanikzai and Shah [80], and Kumar and Kumar Barua [7], who emphasized the critical nature of cybersecurity risks to SCF that systemic failures could lead to a collapse in the financial system. In the same group of economic factors as THR5 and THR10, THR4 - Trade war and economic sanction is also a factor in the cause group and is in the top 10 factors affecting SCF in Vietnam. Previously, Fajgelbaum and Khandelwal [81] also pointed out that trade war and economic sanctions impact trade, import, and export of businesses, thereby affecting the progress of SCF. The US-China trade war in recent years is a typical example. The use of sanctions and economic retaliation between the US and China - the world's two largest economies - causes a huge impact on the global supply chain. The consequences of these sanctions greatly affect the import of raw materials and export of finished products by many Vietnamese businesses, thereby affecting the supply chain and financial supply chain in Vietnam [82]. Thus, the five factors mentioned above are all prominent threats affecting the financial supply chain in Vietnam today.

In addition to factors such as THR9-currency exchange rate fluctuations, THR2-political instability, THR7-inflationary pressures,

THR6-trade disputes, and THR8-transnational commercial crimes, these elements are identified as exerting a comparatively lower impact on the Supply Chain Finance (SCF) than THR5, THR10, THR20, and THR16. Previously, Song et al. [53] pointed out that inflation and exchange rate fluctuations are the factors in the cause group, with a major impact on the financial supply chain. (Giannakis and Papadopoulos [51] also stated that trade instability and economic crimes have a significant impact on supply chain operations and financial supply chains. The aforementioned factors are categorized within the effect group, denoting consequences resulting from the cause factor group. Non-traditional security (NTS) threats in economic, environmental, and technological realms contribute to exchange rate fluctuations, and inflation, impact political stability, and create conditions conducive to the operation of economic criminals. These factors collectively influence both the operational efficiency of the broader supply chain and the specific activities within the financial supply chain.

Analyzing the correlation relationships, the research result has indicated four primary root causes that significantly impact various factors THR5, THR20, THR10, and THR16. Global economic downturn substantially impacts currency exchange rates (THR9) with an influence index of 0.2359, ranking it exceptionally high. Similarly, the influence of the global economic downturn on political instability (THR2) is noteworthy at 0.2389. THR5 emerges as the most pivotal factor, wielding considerable influence over other factors. Global economic downturn causes many impacts on exchange rates, policies, and inflation due to fiscal and monetary policies of countries, while also affecting political stability and people's welfare. The financial crisis also has similar impacts, greatly affecting the supply chain and supply chain finances. The financial crisis manifests a pronounced impact on vital economic sectors, particularly THR7 and THR8. The repercussions extend to exchange rates and economic downturns, as outlined by Lin [83] and Riedel [84] in Vietnam. In other aspects, THR20 delves into health-related issues, a pertinent concern illustrated by the recent global pandemic. Research results have shown that THR20 belongs to the group of cause factors, affecting other factors. This result is also shown by Babu and Yadav [36]. The epidemic affects the global supply chain and the smooth operation of SCF, causing social instability, political instability, and a strong impact on global financial issues. Effect group factors, such as THR9, THR2, THR7, THR6, and THR8, exhibit heightened susceptibility to influence compared to other factors, with their influence levels falling below the average. These factors's influence commented by Klapper and Randall [85] that the impact of these factors on political, finance, economics, and global epidemics is quite low. Those factors are often the consequences of the macro-environmental fluctuations mentioned above. Based on the comprehensive analysis and insights from prior research, this study ranks the top 10 risks in a financial supply chain and delineates their impact relationships.

As found by the Z-Fuzzy Topsis method, sustainable practices, continuous monitoring, and compliance, and risk hedging strategies are the best ways to limit and resolve non-traditional security challenges affecting the financial supply chain. Practices sustainable supply chain practices are essential for restructuring production systems, enabling businesses to proactively manage limited raw material supplies from foreign partners. This approach facilitates the development of interconnected networks within the supply chain, minimizing disruptions and ensuring the stability of SCF [86]. Sustainable SCF is instrumental in enhancing overall supply chain sustainability by fostering relationships and capabilities that contribute to sustainability improvements [87]. In the current context, where various factors negatively impact supply activities, the sustainable implementation of the financial supply chain is crucial for fostering sustainability, flexibility, transparency, and promoting the development of the financial sector. This strategy can approach most of the NTS Threats found above. Continuous monitoring and compliance in supply chain finance are vital for ensuring the integrity, security, and efficiency of financial transactions [57]. Organizations participating in the supply chain must adopt robust practices to minimize risks, enhance transparency, and meet compliance requirements. Compliance is not merely a legal obligation but a strategic imperative, especially in cross-border financial transactions involving multiple parties. Continuous monitoring and compliance also serve as a defense against threats such as political instability (THR2), changes in laws and regulations (THR3), and social unrest (THR11). By minimizing the impacts of THR2, THR3, and THR11, continuous monitoring and compliance emerge as rational and effective solutions to address the challenges posed by non-traditional security threats to the financial supply chain. Hedging strategies in the SCF are one way to regulate relevant trade policies and they will become more effective in limiting financial instability [66]. Risk prevention strategies also help have plans to predict fluctuations that may occur during circulation within the financial supply chain, thereby optimizing cash flow. This measure is also widely used in this approach which aims to reduce the impact of poorly performing assets or unforeseen events on the overall portfolio and as a mitigation risk, helping businesses overcome market fluctuations and uncertainties in SCF. This strategy is especially effective in preventing macroeconomic fluctuations, trade disputes, and predictable technology incidents.

5. Conclusions, implications, limitations, and future work

5.1. Conclusions

The research results have answered the initially posed research questions, specifically as follows.

- (i) Regarding NTS threats affecting SCF in Vietnam, this research has identified nineteen non-traditional security threats significantly impacting Vietnam's supply chain financial in Vietnam. Initially, the study team gathered insights from prior research, revealing twenty-two challenges. However, after conducting expert surveys, three challenges were omitted, resulting in a concise list of nineteen critical issues affecting Vietnam's financial supply chain. Utilizing the DELPHI method, our study probed the interrelations among these identified threats.
- (ii) Regarding the weight of the influence of NTS threats on SCF as well as the cause and effect relationship between threats, eight challenges, namely the trade war and economic sanction, global economic downturn, financial crises, social unrest,

cybersecurity risks, pandemics, health crises, climate change, and natural disasters, along with natural resource scarcity, were categorized as cause factors. The remaining challenges fall under the classification of effect factors. Among the eight causes affecting the supply chain: global economic downturn, financial crises, cybersecurity risks, pandemics, and health crises, four non-traditional security concerns stand out.

- (iii) Regarding propose and ranking solution strategies, this research has systematically examined these issues and proposed ten prioritized strategies to mitigate their impact. These solutions encompass sustainable practices, risk reduction techniques, community involvement, supply chain resilience plans, robust cybersecurity measures, blockchain application, flexible currency management, ethical supply chain practices, and market diversification. Notably, sustainable practices (S10) emerged as the top-ranked solution, followed by continuous monitoring and compliance (S3) and risk reduction strategy (S2) as the subsequent high-rated remedies. These strategic approaches are deemed effective in addressing the complexity of non-traditional security risks. Moreover, S6 proved to be the second most successful option, while (S1) market diversification - was identified as the least effective solution. In summary, this study adeptly tackled three key issues, providing comprehensive insights and proposing actionable remedies for practical implementation.

5.2. Theoretical implications

By incorporating NTS threats, the study expands the theoretical framework of SCF beyond traditional internal threats and operational risks, highlighting the importance of a holistic approach to managing supply chain risks. The study introduces a novel combination of MCDM methods with the Z-number concept and Fuzzy set theory, offering a new approach to dealing with uncertainty and enhancing the accuracy of risk assessments. These methods can be applied in various fields requiring precise risk evaluation under uncertain conditions, contributing to the development of more robust analytical tools in risk management research. Additionally, the use of the Z-Fuzzy DELPHI technique to validate NTS threats identified through a literature review establishes a rigorous methodological framework for identifying and validating critical risk factors, serving as a model for other researchers. The application of the Z-Fuzzy DEMATEL method to determine causal relationships and weight the impact levels of NTS threats enhances the theoretical understanding of threat interactions and their influence on SCF, which can be extended to analyze complex interdependencies in other areas. Furthermore, the employment of the Z-Fuzzy TOPSIS method for ranking proposed strategies offers a systematic approach for prioritizing response strategies, contributing to the theoretical toolkit available for strategic planning and decision-making. Identifying four root cause factors—global economic downturn, pandemic and health crisis, financial crisis, and cybersecurity risk—as having the most decisive influence on SCF provides a theoretical basis for prioritizing these threats in future research and policy-making, leading to more effective risk mitigation strategies.

In theoretical term, this study suggests incorporating NTS threat factors into the supply chain model alongside endogenous threats, thereby enhancing the comprehensiveness of risk assessment. The study's findings that sustainable practices and risk-hedging strategies are the most effective responses to NTS threats underscore the necessity of integrating sustainability and proactive risk management into theoretical models of SCF, highlighting the importance of developing sustainable and resilient economic practices for long-term stability. Finally, emphasizing the importance of macroeconomic, technological, and environmental factors in building sustainable businesses and monitoring economic fluctuations provides a comprehensive theoretical foundation for including these elements in SCF models, guiding future research towards incorporating broader environmental considerations into economic resilience strategies.

5.3. Managerial implications

This study holds substantial managerial implications for both SCF practitioners and the research community. Firstly, by meticulously uncovering and characterizing 19 non-traditional security challenges affecting Vietnamese supply chains, businesses, governments, and stakeholders gain a nuanced understanding of the challenge's intricacies and their potential impacts on financial supply chains and associated networks. Secondly, through a comprehensive examination of correlation, weighting, and ranking of difficulties, the study provides a detailed overview of each challenge's composition and significance. These insights serve as valuable tools for businesses and organizations, enabling them to formulate agile strategies and foresee, prevent, and address security issues while offering unique perspectives on unconventional forms of protection. Ultimately, identifying ten viable solutions contributes to the advancement of supply chain research, benefiting Vietnam and enhancing the global landscape of financial supply chains. These methodologies enhance national and international economic performance and serve as instrumental tools in fortifying the resilience of supply chains financial.

This study offers valuable insights and concrete recommendations to address the escalating complexity within the financial supply chain industry. In summary, as SCF is crucial for the effectiveness and smooth operation of supply chains, this study offers a broader perspective on analyzing potential threats and proposing solutions. This helps stakeholders, including businesses, policymakers, and customers, to mitigate risks and enhance the performance of supply chains and the overall economy.

5.4. Limitations

Despite the efficiency of the research methodologies, users still face challenges due to their complexity, computational demands, and technological intricacies. To address this, developing a user-friendly interface, application, or platform is imperative for ease of accessibility and utilization of the techniques. A notable challenge lies in data reliance, requiring a substantial volume of data for

accuracy, which, given the study's limited sample size, may compromise the precision of the findings. Additionally, the methodology's effectiveness hinges on the quality of expert input, introducing subjectivity and potential bias in evaluations from different experts. Sensitivity Analysis is yet to be incorporated. Notably, the study did not leverage the latest techniques, potentially impacting the outcome. Lastly, the absence of specific evidence from Vietnam, being a novel research area, raises concerns about the authenticity and broad applicability of the findings in the Vietnamese context. Ensuring practical utility and widespread relevance in Vietnam becomes challenging without specific evidence.

5.5. Future research

After thoroughly examining the subject, the research team advocates for additional investigations to overcome identified limitations and refine the model's accuracy and adaptability across diverse contexts. Delving into these recommendations marks a pivotal stride toward more comprehensive and fruitful research endeavors. A fundamental proposition entails expanding the study's scope, potentially by augmenting samples or diversifying data, aiming to heighten accuracy and enrich the dataset for intricate and nuanced modeling. Equally imperative is the augmentation of experts participating in evaluations to ensure the model's applicability and reliability. Leveraging the extensive expertise of these specialists enhances the model's realism and utility in various scenarios. Introducing sensitivity analysis emerges as another instrumental strategy for model enhancement. This widens our comprehension of the model's responsiveness to changes, bolstering its versatility by assessing sensitivity to input variable variations. These recommendations, presented by the research team, offer valuable avenues for further exploration and understanding of the subject in future studies.

Funding

We would like to acknowledge the financial support provided by FPT University Vietnam, through official funding under Decision No. 138/DHFPT/2023.

Availability of data and materials

Data will be made available on request.

CRediT authorship contribution statement

Phi-Hung Nguyen: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Lan-Anh Thi Nguyen:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Formal analysis. **The-Vu Pham:** Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Kim-Anh Nguyen:** Writing – original draft, Visualization, Validation, Software, Resources, Investigation. **Mai-Anh Ngoc Nguyen:** Writing – original draft, Visualization, Validation, Resources, Formal analysis. **Le-Dung Thi Nguyen:** Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation. **Luong-Tuan Nguyen:** Writing – original draft, Visualization, Validation, Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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